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**Evolutionary dynamics shaping accessory
genome hotspots in closely related *E. coli*
ST131 genomes**

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Abstract

Bacterial genomes exhibit substantial structural diversity driven by horizontal gene transfer, recombination, and the activity of mobile genetic elements (MGEs). This diversity is often concentrated in accessory genome “hotspots,” which play a key role in adaptation and the acquisition of traits such as antibiotic resistance. In this thesis, we investigate the evolutionary dynamics shaping these hotspots in 222 closely related *Escherichia coli* ST131 genomes using a pangenome graph framework.

Building on local pangenome graphs constructed with PanGraph, we develop a computational approach to cluster genomic junctions based on homologous recombination signals, reconstruct ancestral genome structures, and identify structural events including insertions, deletions, inversions, and translocations. These events are further characterized through functional annotation and sequence similarity searches to infer their potential origins.

Our results show that homologous recombination frequently contributes to sequence divergence but only rarely introduces structural changes. Instead, MGEs emerge as the dominant drivers of accessory genome variability, accounting for the majority of structural events. We find that insertions occur far more frequently than deletions and are often associated with elements such as insertion sequences and prophages. Additionally, hotspot diversity is not strongly correlated with the number of recombination events, suggesting that local structural complexity arises primarily from repeated MGE activity rather than recombination alone.

Finally, we integrate structural, functional, and evolutionary information into an interactive visualization framework, enabling systematic exploration of accessory genome hotspots. Together, our findings provide new insights into the mechanisms shaping bacterial genome evolution and highlight the central role of mobile genetic elements in driving genomic diversity within clinically relevant lineages such as *E. coli* ST131.

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Chapter 1

Introduction

Bacterial evolution is driven both by the gradual accumulation of mutations through vertical inheritance from parent to offspring [1] and by large-scale genetic changes via horizontal gene transfer (HGT), gene duplication, and gene loss [2, 3]. The three main mechanisms of horizontal gene transfer are the uptake of free DNA from the environment (transformation) [4], bacteriophage-mediated transfer (transduction) and direct cell-to-cell DNA transfer via a conjugative pilus (conjugation) (Fig. 2.1) [3, 5, 6]. Mobile genetic elements (MGEs) play a central role in these processes, as they mediate transduction and conjugation [7, 8]. Consequently, bacterial genomes are highly dynamic, large genomic regions are shared by only a subset of isolates, and even closely related strains can differ substantially in gene content [9, 10, 11]. This extensive variability not only reflects the impact of HGT and gene turnover but also poses considerable challenges for comparative genomic analyses.

The concept of the pangenome, defined as the complete repertoire of genes within a species [12], captures this genomic fluidity. The pangenome is typically partitioned into the core genome, which comprises genes shared by all isolates, and the accessory genome, which includes genes present in only a subset of isolates [12, 13]. While the core genome mainly encodes essential cellular functions, the accessory genome represents a dynamic and variable gene pool that can be gained or lost through horizontal gene transfer [3]. This flexibility enables bacteria to rapidly acquire traits that enhance survival under specific environmental conditions [14]. As a result, genes associated with virulence [15], antibiotic resistance [16], and specialized metabolic functions [17, 18, 19] are frequently located in the accessory genome. By enabling ecological adaptation and providing competitive advantages, the accessory genome influences bacterial fitness, affects host–microbe interactions, and plays an important

role in bacterial evolution.

Evolutionary drivers of accessory genome hotspots

Structural diversity observed in the accessory genome is largely driven by a combination of host-specific recombination mechanisms and the activity of mobile genetic elements (MGEs) [7, 20], both of which are discussed in chapter 2. Rather than being distributed uniformly across the genome, this variability often concentrates in regions of particularly high genetic diversity, commonly referred to as *hotspots* [20, 21].

One important driver of this clustering is the integration behavior of MGEs. When mobile elements insert into essential or adaptive genetic structures, they can impose a substantial fitness cost on the host. Consequently, many MGEs preferentially target genomic sites where their insertion is less disruptive. A common example is the frequent insertion of integrases into tRNA or tmRNA genes [22]. Over time, repeated insertions and recombination events at these permissive loci give rise to highly variable regions within the pangenome [20].

These hotspots often accumulate genes that provide adaptive advantages to the host. A prominent example is the formation of defense islands, genomic regions enriched in anti-phage defense systems [23, 24]. Bacteria employ diverse defense mechanisms against phages, including restriction-modification systems and CRISPR-Cas systems that target viral nucleic acids, abortive infection systems that induce host suicide, as well as other mechanisms such as chemical defense and signaling systems. Because maintaining all possible defense systems would be costly, bacteria often rely on horizontal gene transfer to acquire defenses that are beneficial under specific environmental conditions [25]. As a result, defense systems are particularly relevant when studying genome diversity and horizontal gene transfer.

However, not all hotspots appear to originate from MGE insertions. In some regions, no identifiable MGEs, integrases, or associations with tRNA or tmRNA genes can be detected, suggesting an alternative mechanism. In these cases, the high sequence diversity observed in the flanking core genes indicates that homologous recombination likely drives diversity at these loci (Figure 1.1d) [9, 20]. Hotspots in general, and especially those associated with homologous recombination, appear more frequently in naturally transformable bacteria, suggesting that recombination plays an important role in genome diversification in these species. Moreover, homologous recombination at hotspots can provide an efficient mechanism for removing elements

with deleterious fitness effects, which may help explain the elevated recombination rates observed in these regions [20].

Pangenome graph representations of bacterial genome diversity

Due to the extensive structural diversity across bacterial genomes, comparative analyses based on a single reference genome or conventional multiple sequence alignments are usually limited to the core genome. Although this approach enables the inference of phylogenetic relationships [6], it overlooks important biological functions encoded within the accessory genome [26, 27]. Pangenome graphs address this limitation by representing both core and accessory genome regions within a unified graph structure. Instead of aligning entire genomes linearly, they partition genomes into shared sequence segments. These segments form nodes in the graph, and each genome is represented as a path that traverses the nodes corresponding to its specific sequence composition and order [28].

Several tools are available for constructing prokaryotic pangenome graphs. Many widely used approaches, including Panaroo [29], PanX [30], Roary [31], PIRATE [32], PanOCT [33], PGAP [34], and PPanGGOLiN [35], define alignable segments at the gene level. These methods rely on annotated genomic features and characterize structural variation primarily in terms of gene presence and absence across genomes. Notably, PanOCT was the first to complement sequence homology with gene neighborhood information to improve the inference of orthologous relationships [33].

In contrast, PanGraph [36] constructs pangenome graphs directly from nucleotide sequences, without relying on gene annotations or predefined gene boundaries. It formulates graph inference as a progressive multiple-genome alignment problem, as illustrated in figure 1.1a. Initially, one graph is created for each genome, representing the entire isolate genome as a single node. Genomes are then aligned iteratively along a guide tree, during which their corresponding graphs are progressively merged. At each merging step, homologous regions are identified and combined into shared nodes, referred to as *blocks* or *pancontigs*. In the final graph, each node corresponds to a multiple-sequence alignment of the respective genomic region. Nodes are directed and store strand orientation information for every sequence they contain. Adjacency relationships observed in individual genomes are encoded by edges con-

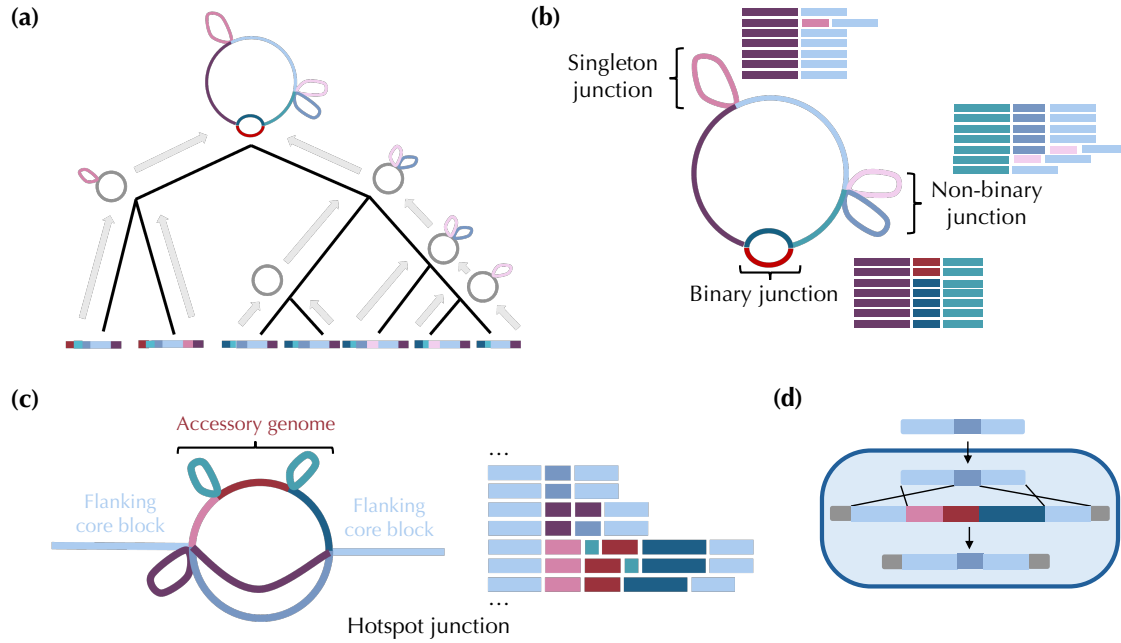


Figure 1.1: Pangenome graph construction and junction structure (adapted from [6, 36, 21]). (a) Illustration of pangenome graph construction with PanGraph using simplified bacterial genomes. (b) Junctions in the graph can be classified into binary and more complex junctions, with binary junctions further divided into singleton and non-singleton junctions. They can be visualized either by their graph representation or by their block structure. (c) Junctions typically consist of two flanking core blocks enclosing accessory genomic regions. If they exhibit particularly high accessory genome diversity, they are referred to as hotspot junctions. (d) Homologous recombination is one of the drivers of the diversity observed in hotspots, as it can result in the exchange of the enclosed genetic material between the two flanking core blocks involved in the recombination.

necting the nodes. By operating directly at the nucleotide level, PanGraph captures structural variation beyond gene definitions and avoids biases introduced by annotation errors. At the same time, it separates nucleotide-level polymorphisms within nodes from larger-scale genomic rearrangements reflected in the graph topology. Because this approach models genomes at high resolution, it is primarily suited for sets of closely related bacterial isolates, although even in such datasets the resulting graphs can become complex.

Within pangenome graphs, the organization of the core genome provides an important structural reference. Rearrangements of the core genome are typically subject to strong selection pressure, which results in the order, or synteny, of core genes being conserved over large evolutionary distances [37, 38]. This conservation allows core genome blocks to serve as a stable reference frame for analyzing diversity in the accessory genome between them, which appears as alternative paths in the graph. Such regions, defined by a pair of consecutive core blocks and the acces-

sory sequence between them, are referred to as core genome *junctions* (Figure 1.1c). Since accessory sequences may be homologous to distant regions elsewhere in the graph, pangenome graphs may become substantially fragmented. To mitigate this issue and improve alignment accuracy, PanGraph can be rerun separately on each junction, producing refined local pangenome graphs. Within these local graphs, the level of diversity can be characterized by the number of alternative paths and the total length of the accessory sequence. Junctions exhibiting particularly high diversity are referred to as *hotspot junctions* (Figure 1.1c) [21]. These concepts provide a framework for studying genome structural variation in bacterial populations. In the following section, they are applied to the analysis of genome diversity in *Escherichia coli* ST131.

Genomic evolution in *Escherichia coli* ST131

The Gram-negative bacterium *Escherichia coli* is one of the most extensively studied model organisms [9]. It occupies a wide range of ecological niches, including the gastrointestinal tract of mammals and birds where it typically exists as a commensal, but it can also act as a pathogen. In addition, *E. coli* is frequently found in host-independent environments such as water, sediments, and soil. Adaptation to these diverse habitats has resulted in a large and flexible gene repertoire [39, 11].

A key feature of *E. coli* is its open pangenome, which continuously gains and loses genes through intra- and interspecies horizontal gene transfer [39] and frequently recombines [40]. As a consequence, genomes that differ by only a few substitutions in their core genome can display substantial variation in accessory gene content and associated phenotypes. This pattern indicates high rates of horizontal gene transfer and highlights the importance of considering the entire genome when studying *E. coli* populations [9, 11, 41]. The *E. coli* genome is therefore highly diverse, with roughly half of the average genome shared among most isolates, while the pangenome is substantially larger [9, 10, 42, 43]. Despite this extensive accessory genome diversity, the synteny of the core genome remains highly conserved [9, 21].

Under natural conditions, *E. coli* is generally not transformable. Transformation can occur only under extreme conditions, which are typically achievable in laboratory settings [44]. Consequently, conjugation and transduction represent the primary routes of horizontal gene transfer in this species. Recombination into the *E. coli* genome can occur through most of the mechanisms described above [45, 46, 47, 48], with the exception of non-homologous end joining [49].

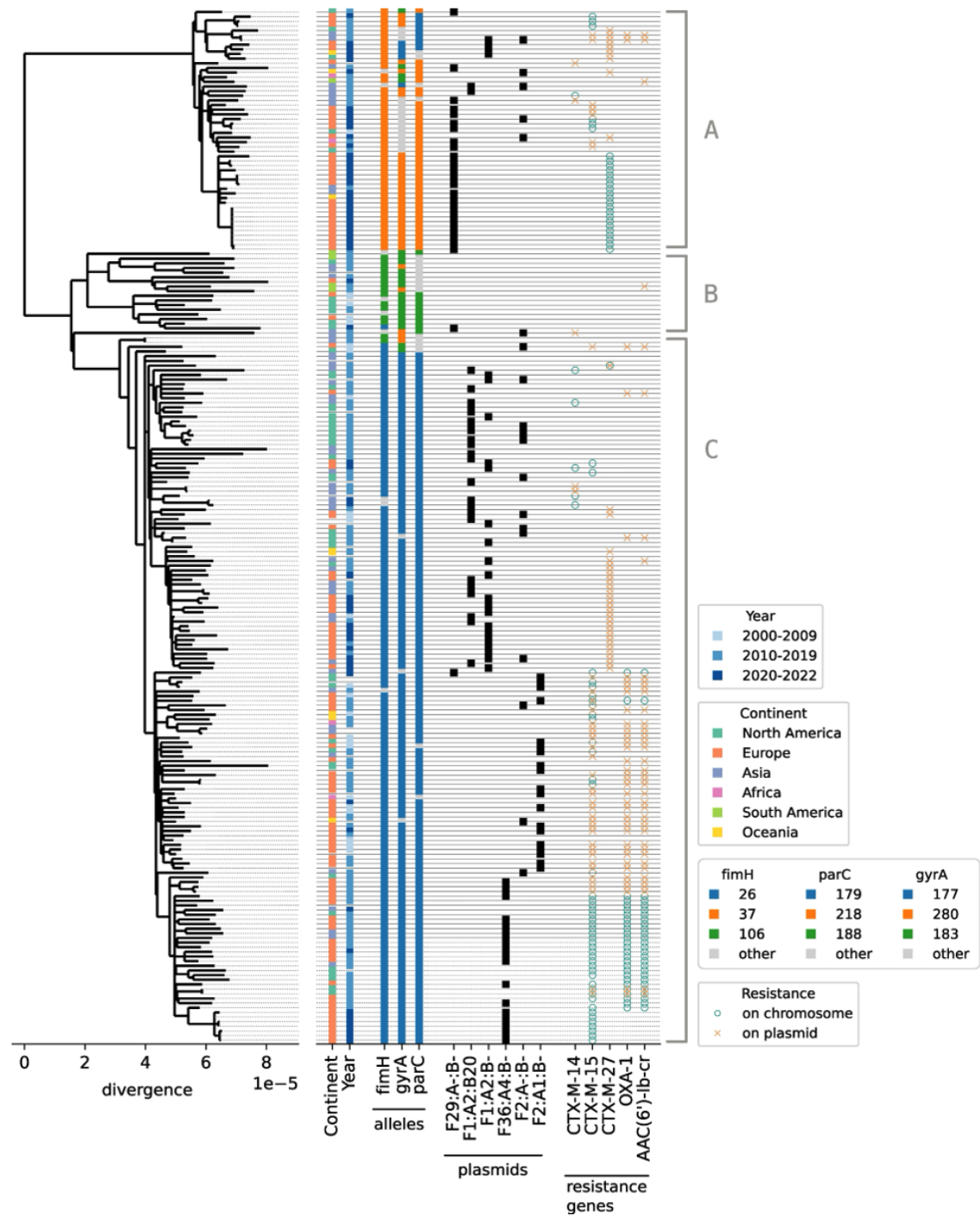


Figure 1.2: Core genome phylogeny and genomic annotations of 222 *E. coli* ST131 isolates (reproduced from [21]). The phylogenetic tree was inferred from the core genome after excluding regions likely affected by recombination (identified by high SNP density). Annotations shown on the right indicate the geographic origin and year of isolation, as well as the allelic variants of *fimH*, *gyrA*, and *parC*. Information on IncF plasmid types and the presence of selected antibiotic resistance genes is also displayed. The phylogeny reflects the global population structure of *E. coli* ST131, with isolates belonging to the three major clades A, B, and C.

Within *Escherichia coli*, the sequence type ST131 has attracted particular attention due to its major clinical relevance. ST131 is a multidrug-resistant lineage that contributes substantially to antibiotic-resistant urinary tract and bloodstream infections worldwide through the expression of extended-spectrum beta-lactamases [50, 51, 52]. The most recent common ancestor of ST131 is estimated to have existed about 100 years ago (TMRCA 1901, 95% highest posterior density interval (HPD) 1842–1948) [51]. However, most clonal expansion occurred during the past 40 years following the widespread introduction of beta-lactam antibiotics [53, 54].

This rapid and recent expansion makes ST131 a suitable system for studying short-term bacterial genome evolution, in which evolutionary patterns are not dominated by frequent recombination [40]. Molari et al. therefore analyzed a collection of closely related ST131 genomes, focusing on a regime where homologous recombination has only a minor influence and can be treated as a perturbation. Using PanGraph, they quantified the evolutionary dynamics of genome structure and content within this lineage, providing the foundation for the analyses presented in this study.

Their dataset comprises 222 high-quality genomes collected from RefSeq, identified as ST131 by multi-locus sequence typing, and closely related to the ST131 reference genome *NC_013654.1* (Mash distance < 0.008 [55]). To construct a core genome alignment, regions with high single nucleotide polymorphism (SNP) diversity were strictly excluded [56], as these regions likely arise from recombination [40] and could distort phylogenetic inference. The resulting phylogeny (Figure 1.2) recapitulates the known global diversity of ST131, including representatives of the three major clades A, B, and C [52]. Its total branch length corresponds to 6040 SNPs across the filtered core genome. The average pairwise divergence is of the order of 74 mutations per Mbp, equivalent to less than 10^{-4} substitutions per site [21].

Beyond core genome variation, Molari et al. identified 519 genomic junctions exhibiting accessory genome variation. These junctions correspond to at least 2000 structural genome modifications occurring at roughly one third the frequency of core genome SNPs. Among these, 328 junctions were classified as *binary*, meaning that only two alternative genomic paths are observed. Eight junctions were excluded because they flank synteny breaks and are not present in all genomes. The remaining 320 binary junctions comprise 273 *singletons*, where the minority path occurs in only one isolate, and 47 *nonsingletons* (Figure 1.1b). Across binary junctions, gains occur roughly ten times more frequently than losses, largely driven by insertions of

insertion sequences and prophages.

The other 191 junctions show higher diversity, with several representing genomic hotspots. These regions often contain multiple mobile genetic elements and frequently harbor bacterial defense systems [21]. This observation is consistent with earlier studies identifying genomic hotspots enriched in defense systems, typically associated with repeated insertions of mobile genetic elements and shaped by homologous recombination [9, 20, 23]. While Molari et al. provide a general characterization of these hotspot junctions, the present study aims to analyze their structural organization and genetic composition in greater detail.

Motivation and Objectives

Although much is known about the mechanisms and diversity of horizontal gene transfer, reconstructing the chronological ordering and rates of the structural events that shape bacterial genomes remains challenging. There are two main difficulties underlying this problem. First, structural changes such as insertions, deletions, inversions, translocations, and recombination-driven exchanges are difficult to identify, particularly in highly variable accessory genome regions. Secondly, placing these events on a phylogeny is complicated by homologous recombination, which can obscure genealogical relationships [40] and thereby hinder the estimation of evolutionary rates.

The second challenge can be mitigated by focusing on an appropriate dataset. Closely related genomes, such as those within the *E. coli* ST131 lineage, provide a regime in which structural changes accumulate gradually and are less likely to overwrite one another, while patterns of vertical inheritance are not yet strongly obscured by homologous recombination.

The first challenge, identifying and characterizing structural events, remains to be solved. The high variability of the accessory genome prevents straightforward alignment-based comparisons, as genomic segments can differ in order, orientation, and copy number. PanGraph addresses this challenge by constructing pangenome graphs that enable comparison of genomic segments across multiple genomes and allow separate analysis of nucleotide-level variation and large-scale structural rearrangements through a block-based representation. Nevertheless, systematically identifying and classifying structural events within these graphs remains difficult, particularly in the presence of duplications, inversions, and translocations. Recent work by Molari et al. has begun to quantify rates of structural evolution in *E. coli*,

focusing mainly on junctions with limited structural variation [21]. In this study, we extend this approach by investigating the dynamics of structural events within highly diverse hotspot junctions across 222 closely related *E. coli* ST131 genomes.

Building on the local junction graphs generated by Molari et al. [21], we analyze each hotspot junction individually. First, we detect homologous recombination events and cluster hotspot junctions accordingly. For each cluster, we reconstruct the most likely structure of the most recent common ancestor and use it as a reference to identify structural events. This enables us to estimate rates of structural evolution. We further characterize the genetic content of hotspot junctions through functional annotation using tools such as geNomad [57], ISEScan [58], IntegronFinder [59], and DefenseFinder [60]. To investigate the potential origins of genomic sequences, we employ fast similarity search tools such as LexicMap [61] and compare results against the large bacterial reference collection AllTheBacteria [62]. Finally, we integrate structural information, functional annotations, and inferred sequence origins into an interactive dashboard, providing a comprehensive overview of hotspot junctions and new insights into the evolutionary dynamics shaping these regions.

Chapter 2

Biological Background

Bacterial recombination mechanisms

Recombination in prokaryotes is commonly classified into *RecA-mediated homologous recombination*, *RecA-independent recombination*, and *site-specific recombination* (Figure 2.1) [6, 45].

RecA-mediated homologous recombination

Homologous recombination (HR) plays a central role in DNA repair, particularly in the repair of double-strand breaks. It restores damaged DNA by using a homologous sequence as a template, which can originate either from a sister chromosome or from homologous foreign DNA present in the cell [63]. When foreign DNA serves as the template, HR is directly linked to horizontal gene transfer, an important driver of bacterial genetic diversity and adaptation [64].

Homologous recombination is initiated through two main pathways, the RecBCD pathway, which primarily repairs double-strand breaks, and the RecFOR pathway, which repairs single-stranded gaps [46]. Both pathways rely on the RecA protein to recognize homologous sequences and catalyze strand exchange [63]. Successful recombination requires sufficiently long homologous regions at both ends of the incoming DNA, typically between 25 and 200 bp [5]. Recombination efficiency declines rapidly as sequence divergence between donor and recipient increases and is generally limited to sequences with less than about 30% divergence [65, 66]. When these requirements are met, donor DNA can integrate into the chromosome. This may include mobile genetic elements, which can either be transported as cargo within

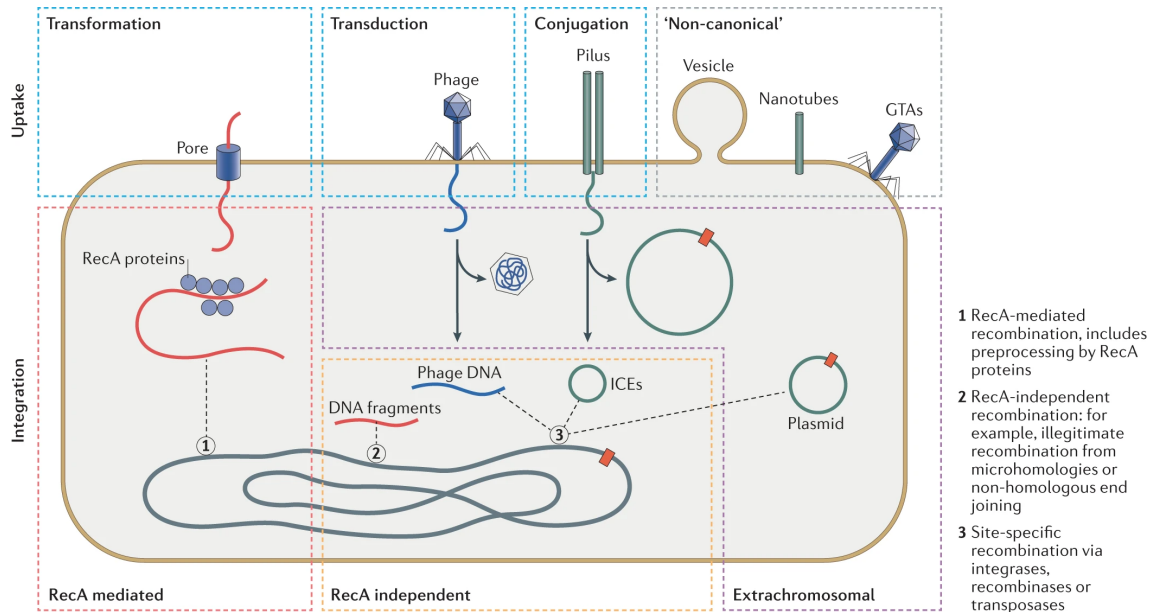


Figure 2.1: Overview of DNA uptake and integration mechanisms in horizontal gene transfer (reproduced from [6]). Horizontal gene transfer begins with the uptake of extracellular DNA, which typically occurs through one of three principal mechanisms: transformation (uptake of free DNA), transduction (bacteriophage-mediated transfer), or conjugation (direct cell-to-cell transfer via a conjugative pilus). In addition, several non-canonical mechanisms have been described, including DNA transfer via membrane vesicles, pilus-like structures called nanotubes, or phage-like gene transfer agents (GTAs). Following uptake, the incoming DNA must integrate into the recipient genome. If sufficient sequence homology exists, integration can occur through RecA-mediated homologous recombination. Alternatively, illegitimate (RecA-independent) recombination requires little or no sequence homology but depends on physical proximity of DNA molecules and includes mechanisms involved in DNA break repair or associated with replication. Mobile genetic elements frequently mediate integration through site-specific recombination using specialized recombinases or transposases.

homologous flanking regions or integrate through their own homology with regions in the host genome where related mobile genetic elements already exist [6].

However, DNA fragments are often degraded before recombination, both in harsh extracellular environments during transformation and intracellularly after uptake, for example by restriction–modification systems. As a result, shorter DNA fragments are more frequently incorporated, increasing the likelihood that recombination removes existing sequences rather than inserting new ones and thereby contributing to the deletion bias observed in bacterial genomes [6, 67].

RecA-independent recombination

RecA-independent, or *illegitimate recombination*, occurs less frequently than homologous recombination [68]. Because it typically requires local proximity of recombining sequences, it mainly results in local changes in chromosomal structure [45]. It

generally involves little or no sequence homology. One example of RecA-independent recombination is non-homologous end joining (NHEJ), a mechanism that repairs double-strand breaks without requiring homologous sequences and has been identified in some, but not all, prokaryotes [69, 49].

Other mechanisms rely on short homologous repeats, including slipped-mispair and single-strand annealing [70]. Slipped-mispair can occur when DNA replication temporarily pauses, allowing the newly synthesized strand to dissociate and reanneal at a nearby repeat sequence. When replication resumes, this can lead to conversion events or to deletions or duplications of a repeat and the intervening sequence. Single-strand annealing represents another pathway for repairing double-strand breaks. In this process, exonucleases degrade DNA ends until complementary single-stranded regions are exposed. These regions can then anneal and ligate, leading to deletion within the resulting sequence [45].

Site-specific recombination

Site-specific recombination is widely used by mobile genetic elements to integrate into host chromosomes and is mediated by specialized enzymes such as transposases and site-specific recombinases [6].

Transposases catalyze the movement of discrete DNA segments between non-homologous insertion sites [47]. Two principal modes of transposition exist, non-replicative and replicative transposition. In non-replicative transposition, the element is excised from its original location and inserted at a new site, leaving a double-strand break at the donor locus that may be repaired by homologous recombination, degraded, or occasionally resealed. Replicative transposition instead duplicates the element and inserts the copy at the target site. If the target lies on the same DNA molecule, deletions or inversions may occur, whereas insertion into a separate molecule produces a cointegrate in which donor and target are joined by two copies of the element. Subsequent recombination resolves this intermediate, leaving each molecule with one copy of the transposable element [71]. Target site selection varies widely and can depend on sequence motifs, local DNA structure, host protein binding, or DNA supercoiling. Preferred insertion sites are often abundant, relatively conserved, and minimize detrimental effects on the host genome [47].

Site-specific recombinases fall into two major families defined by their cleavage mechanism, serine recombinases and tyrosine recombinases, the latter often referred to as integrases [48, 72]. These enzymes bind two recombination sites, pair them,

cleave and exchange DNA strands, and rejoin the products [48]. Depending on the relative orientation and location of the sites, recombination can generate inversions, deletions, or integrations. Intramolecular recombination produces inversions when sites are in head-to-head orientation and deletions when arranged head-to-tail, whereas recombination between different DNA molecules leads to integration. Unlike transposition, the recombination sites are predefined (typically 20–200 bp) and the reaction is reciprocal, requiring neither DNA synthesis nor degradation [71, 48]. Many integrases preferentially target tRNA or tmRNA genes, which explains why mobile genetic elements are frequently found adjacent to these loci [22, 20].

Mobile genetic elements

Mobile genetic elements (MGEs) are DNA segments capable of actively moving between genomes. They encode proteins that enable their movement either within genomes (intracellularly) or between bacterial cells (intercellularly). Through this mobility they can strongly influence microbial adaptation, although their interests are not necessarily aligned with those of their hosts [8]. MGEs use diverse mechanisms to spread between cells and to integrate into host genomes, relying on the recombination processes described above.

Two of the most common mechanisms of intercellular transfer are conjugation and transduction. Conjugation requires direct cell–cell contact and is mediated by conjugative plasmids and integrative conjugative elements, which transfer themselves through a pilus [7, 8]. In transduction, bacteriophages occasionally package bacterial DNA instead of viral DNA, producing transducing particles that inject this DNA into new host cells during infection. Three main forms of phage-mediated transfer are known. In generalized transduction, phages mistakenly package random bacterial DNA fragments by recognizing pseudo-packaging motifs. In specialized transduction, incorrect excision of a prophage results in the transfer of bacterial genes adjacent to the phage integration site. Both processes occur at relatively low frequencies [73]. In contrast, lateral transduction can transfer large chromosomal regions at much higher frequencies [74]. This process involves multiple prophages integrated into the host genome, enabling stepwise DNA packaging in which some prophages remain integrated and serve as substrates for packaging while others have already excised. Packaging starts at the phage genome and continues into adjacent DNA, which typically contains other prophages but can also include bacterial DNA. Through these mechanisms, phage transduction appears to be the major route of

bacterial gene transfer [73], even between different species [15]. Alternatively to active transfer mechanisms, mobile genetic elements can also spread by hitchhiking within any horizontally transferred DNA sequence [75].

In the following, I introduce several common types of mobile genetic elements, which are summarized in Figure 2.2.

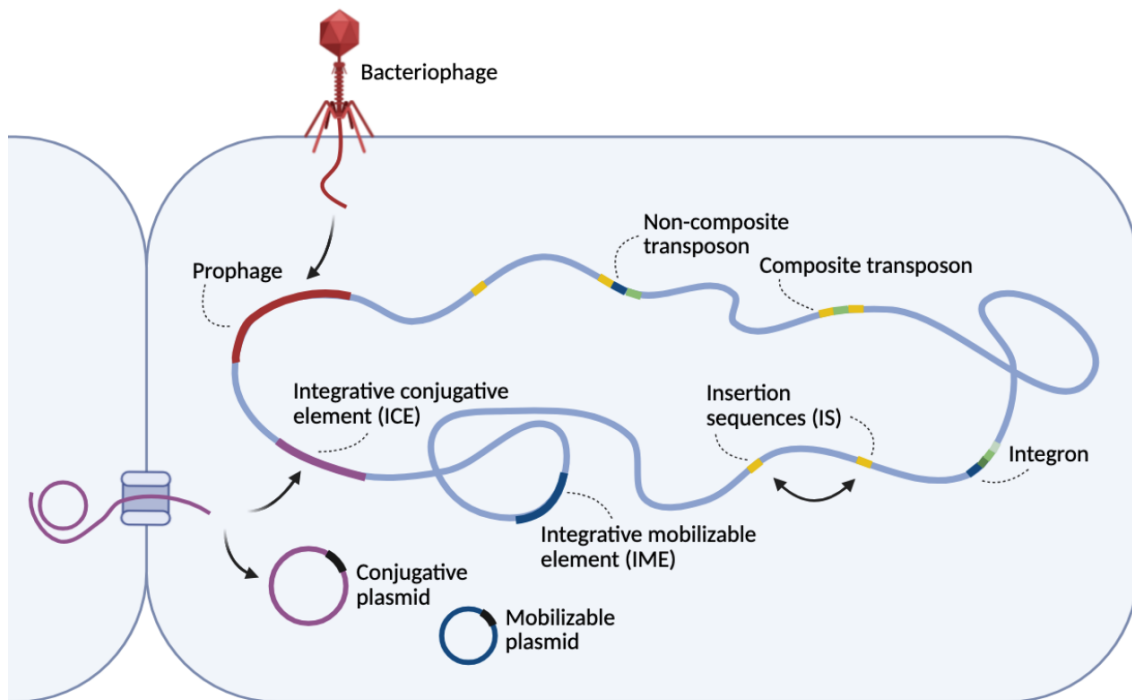


Figure 2.2: Overview of common types of mobile genetic elements. Bacteriophages are viruses that infect bacteria by injecting nucleic acids into the host cell, which may integrate into the bacterial chromosome as a prophage. Conjugative plasmids and integrative conjugative elements (ICEs) encode the machinery required for conjugation and can transfer between cells via a pilus. Related elements, such as mobilizable plasmids and integrative mobilizable elements (IMEs), lack a complete conjugation system and therefore rely on the conjugation machinery of other elements for transfer. While plasmids typically replicate autonomously and remain extrachromosomal, integrative elements are inserted into the chromosome through site-specific recombination. There are many different types of transposable elements, which are generally characterized by using transposases to move. Insertion sequences (IS) are the smallest autonomous transposable elements and usually encode only a transposase required for their mobility. Larger transposons can occur as composite transposons, in which accessory genes are flanked by two insertion sequences that mediate their transposition, or as non-composite transposons that generally transpose replicatively and encode recombinases to resolve cointegrate intermediates. Finally, integrons act as gene capture and expression platforms that integrate gene cassettes and facilitate the accumulation and expression of adaptive genes.

Bacteriophages

Bacteriophages are viruses that infect bacteria. They consist of a protein capsid enclosing nucleic acids and can integrate into bacterial genomes as prophages,

where they exploit host replication machinery and contribute to the spread of genetic material. Following infection, phages can adopt either a lysogenic or a lytic lifecycle. In the lysogenic state, the phage genome is maintained within the host and replicated alongside it, whereas in the lytic cycle, phages replicate independently and ultimately lyse the host cell, releasing new viral particles and potentially transferring genetic material to other bacteria. Temperate bacteriophages, which predominantly rely on the lysogenic lifestyle, often carry genes that are beneficial to their host, such as defense systems against other phages. By maintaining a long-term association with the host, these elements align their evolutionary success with host survival. Phages are highly diverse and are thought to be the most abundant biological entities on Earth [76, 7, 77]. Their genomes typically range from about 3 to 200 kbp, although jumbophages can reach up to 500 kbp [78, 79]. Integration usually occurs via site-specific recombination mediated by recombinases, although transposable phages using replicative transposition also exist [80].

Conjugative plasmids and integrative conjugative elements

Conjugative plasmids and integrative conjugative elements (ICEs), also known as conjugative transposons, encode the machinery required for conjugation as well as accessory genes. Plasmids are typically circular extrachromosomal DNA molecules, whereas ICEs integrate into the host chromosome via site-specific recombinases [7, 8, 64]. Both element types vary widely in size, ranging from roughly 1 kbp to several hundred kbp [81, 82]. Plasmids are characterized by their ability to replicate independently of the chromosome [5], whereas ICEs are generally considered incapable of autonomous replication. However, recent studies have identified ICEs capable of autonomous replication [83], as well as plasmids integrated into host chromosomes, suggesting a high similarity between the two element types [8].

Mobilizable plasmids and integrative mobilizable elements

Mobilizable plasmids and integrative mobilizable elements (IMEs), also known as mobilizable transposons, resemble their conjugative counterparts but lack a complete conjugation machinery. They therefore rely on conjugative elements for transfer between cells [84] and are typically smaller. Despite this, they appear to be at least as abundant as conjugative elements [8].

Composite and non-composite transposons

Composite and non-composite transposons often carry accessory genes that benefit their bacterial hosts, including genes conferring antibiotic or heavy metal resistance, phage resistance, metabolic capabilities, vitamin synthesis, antimicrobial compound production, or nitrogen fixation. Both types are flanked by terminal inverted repeats [64, 85]. Composite transposons typically range from 2.5 to 60 kbp and consist of accessory genes flanked by two similar or identical insertion sequences, at least one of which encodes a functional transposase that mediates transposition, usually via a non-replicative mechanism [64]. Non-composite transposons are usually 3–50 kbp long [86]. They undergo replicative transposition and encode site-specific recombinases that resolve the resulting cointegrated intermediate [87].

Insertion sequences

Insertion sequences (IS) are the smallest and most abundant autonomous transposable elements in prokaryotic genomes, typically ranging from 400 to 10,000 bp. They generally encode a transposase responsible for their mobility [75, 58]. IS elements are classified into families mainly based on transposase sequence, as well as characteristics such as sequence similarity, target sites, and open reading frame organization. Originally defined by the absence of cargo genes, this distinction from transposons is now considered less strict [75].

Integrations

Integrations are genetic platforms that enable bacteria to capture and express gene cassettes, promoterless genetic elements that can be converted into functional genes. In this way, they play a major role in bacterial adaptation and the acquisition of multidrug resistance. Integrations encode a tyrosine recombinase within their *intI* gene that mediates RecA-independent integration of new gene cassettes. They also contain a primary recombination site (*attI*) and a Pc promoter that drives expression of integrated cassettes.

Integrations themselves are stable and immobile but can become mobile when embedded in transposons or plasmids, forming mobile integrations that can spread between bacteria. In contrast, chromosomal superintegrations are large integration structures embedded in bacterial chromosomes that contain extensive arrays of gene cassettes and mainly serve as reservoirs of genes for long-term genome diversification. Their size varies with the number of integrated gene cassettes and typically ranges from a few

kilobases in small mobile integrons to up to about 200 kbp in superintegrons [88, 89].

Chapter 3

Methods

This study builds on the dataset introduced by Molari et al. [21]. The dataset comprises 222 complete *E. coli* ST131 genomes retrieved from RefSeq [90, 91] and selected based on sequence typing and genetic similarity criteria. The genomes were subjected to quality control, after which a pangenome graph was reconstructed using PanGraph. A phylogeny was inferred from a recombination-filtered core genome alignment, excluding regions with high SNP density. Subsequently, loci exhibiting accessory genome variation, termed junctions, were identified. For each junction, a separate pangenome graph was reconstructed, resulting in 518 junctions that retained accessory genome diversity. Each local junction was functionally annotated using DefenseFinder [60], ISEScan[58], geNomad [57], and IntegronFinder [59] to identify defense systems, insertion sequences and their corresponding IS families, prophages, and integrons. Gene annotations were obtained from the corresponding RefSeq genome assemblies. Detailed descriptions of dataset retrieval, quality control, pangenome graph construction, functional annotation, and core genome phylogeny inference are provided in the supplementary material of Molari et al. [21].

Inference of junction evolution and structural genomic changes

To structurally analyze accessory genome junctions, we developed a computational framework to cluster similar accessory paths in a junction, infer ancestral block structures, and detect structural genome events (Figure 3.1a-d).

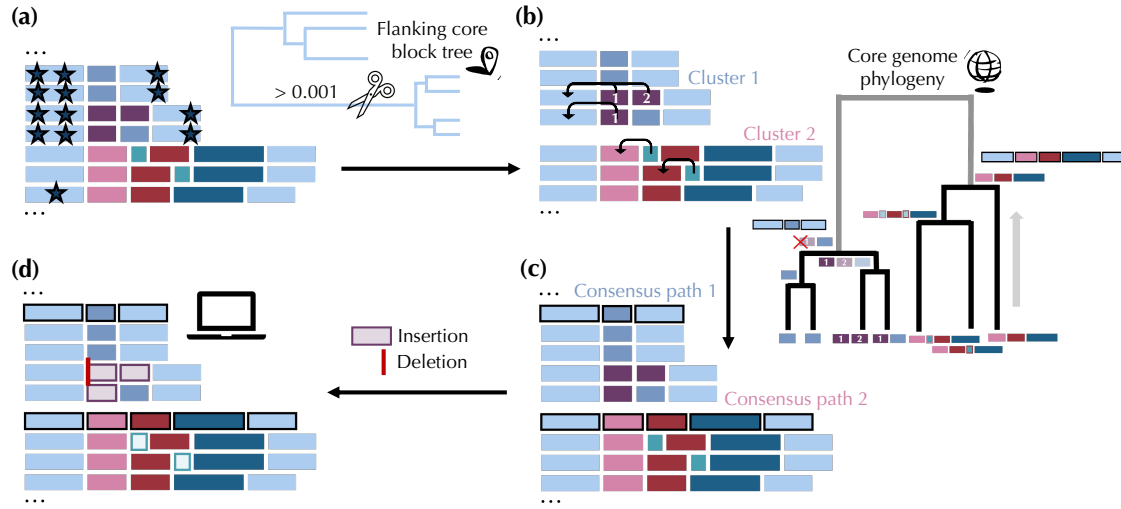


Figure 3.1: Computational framework for junction clustering, ancestral block reconstruction, and detection of structural events. (a) Junctions are clustered based on sequence similarity of their flanking core genome blocks. For each junction, a local phylogeny is inferred from the combined alignment of its core blocks. Branches exceeding 0.001 substitutions per site are cut, and clusters correspond to the remaining connected clades. (b) For each cluster, the ancestral block structure is reconstructed. Duplicated blocks are resolved by selecting an anchor block to the left, and remaining ambiguities are handled by assigning numerical labels to ensure unique block-context combinations. The presence or absence of deduplicated blocks in the most recent common ancestor is then inferred along the global core genome phylogeny using the Fitch algorithm. If the root state is ambiguous, gain and loss events are distinguished using block sequence similarity. Single blocks are interpreted as single insertion events. (c) Consensus paths are used as references to detect structural events such as insertions, deletions, translocations, and inversions. (d) The final output summarizes the evolutionary history of each junction, separating homologous recombination from secondary structural variation. The complete history can be explored in an interactive dashboard.

Grouping junctions by underlying homologous recombination events

Homologous recombination can introduce large structural changes in the accessory genome, resulting in a block structure that appears clustered by overall architecture and can obscure the detection of subsequent, smaller-scale events. Therefore, junctions must first be grouped according to the homologous recombination events that might have contributed to their diversity before meaningful statements about finer structural variation can be made. To achieve this, we exploit the observation that homologous recombination events are typically associated with increased sequence divergence in the flanking core blocks [20], a pattern that is also evident in our data. We therefore cluster junctions based on sequence similarity in these flanking core regions. First, the combined core blocks from all isolates are aligned using MAFFT v7.526 [92, 93] with default parameters and automatic algorithm selection. Phylo-

genetic trees are then reconstructed from the resulting alignment using maximum-likelihood inference implemented in `FastTree 2.2.0` [94], under the General Time Reversible (GTR) substitution model for nucleotide sequences, with gaps treated as missing data. Clustering is performed by cutting the resulting trees at branches exceeding 0.001 substitutions per site (Figure 3.1a), a threshold substantially higher than expected under vertical descent ($< 10^{-4}$), as observed in the phylogeny based on the recombination-filtered core genome alignment.

Phylogeny-based reconstruction of consensus paths

To reconstruct consensus paths representing the most likely block structure of the most recent common ancestor (MRCA) within each cluster, we leverage isolate relationships inferred from the core genome tree, which reliably approximates the clonal phylogeny of closely related isolates [6].

We first resolve duplicated blocks. For each duplicated block, context anchors are defined as the nearest blocks to the left that (i) occur in more than 20% of isolates in the cluster, (ii) are not duplicated, and (iii) are not inverted. To prevent long duplications or inversions from interfering with anchor identification, isolates are excluded if fewer than 60% of their blocks can serve as anchors, provided that at least 80% of isolates remain after filtering. When this condition is not met, the filtering threshold is relaxed such that only isolates with fewer than 30% of anchor blocks are removed, or no isolates are excluded. If block–anchor combinations remain non-unique due to repeated blocks in close proximity, the context is further distinguished by appending an index reflecting the order of the blocks along the genome.

After deduplication, each remaining isolate is binary encoded to indicate the presence or absence of every block ID–context combination. Blocks occurring in the same context but with opposite strandedness are treated as identical to avoid inversions artificially enforcing absence in the consensus. Block presence or absence is inferred using the Fitch algorithm [95] applied to the binary-encoded sequences along the core genome tree, subsampled to the remaining isolates of the cluster. When the root state is uniquely determined, it defines the consensus state. Remaining ambiguities are resolved using sequence similarity between different block occurrences within the cluster. Blocks with low sequence similarity (average pairwise divergence per site > 0.001) are interpreted as independent insertion events and are therefore considered absent in the MRCA, whereas highly similar blocks are assumed to have been present in the MRCA and subsequently lost in some lineages. To calculate the

average pairwise divergence within a block, all block occurrences within a cluster are aligned using **MAFFT v7.526** [92, 93], applying the same parameters used for the core genome block alignments. If a block occurs only once, sequence similarity cannot be evaluated, and a single insertion event is assumed, implying absence in the MRCA (Figure 3.1b).

Finally, the consensus path is reconstructed by removing all non-consensus blocks from the isolates and deriving a majority path from the filtered block arrangements. After consensus construction, both consensus paths and isolates are re-deduplicated while considering all isolates to ensure consistent and unique block ID–context combinations. Because all clusters are included at this stage blocks require only a minimum frequency of 10% to qualify as anchors.

Detecting structural events relative to the consensus path

Structural events are detected by comparing each isolate to the reconstructed consensus path (Figure 3.1c). We distinguish four event types: insertions, deletions, inversions, and translocations. The context definition used for consensus reconstruction excludes blocks that are duplicated or inverted in any isolate. In the presence of large duplications or inversions, this can lead to extended regions without context anchors, making the definition unsuitable for detecting smaller structural events. Therefore, instead of relying on a global context definition, each isolate is compared individually to the consensus path, and context anchors are defined separately for each isolate. For a given isolate, context anchors are blocks that occur exactly once in both the consensus and the isolate path and share the same strandedness. We then perform a preliminary detection step to identify translocations, since such blocks should not serve as anchors. The isolate is iteratively scanned for translocated block IDs, which are removed from the set of potential anchors, and the context definition is updated accordingly. This procedure is repeated until no additional translocated blocks are detected. Using the resulting final context definition, structural events are classified as follows:

- **Translocations** are blocks that share the same block ID but occur in a different context compared to the consensus path.
- **Insertions** are blocks that do not occur in the consensus path with the same block ID and context and are not classified as translocations.
- **Deletions** are blocks present in the consensus that do not appear in the isolate

with the same context and are not classified as translocations.

- **Inversions** are blocks that appear in the same context as in the consensus but with opposite strandedness.

If identical events occur in adjacent nodes, they are merged into a single event. Inversions are treated as uninterrupted by insertions or translocations. Because the analysis is restricted to the local context of a single junction, we distinguish insertions from translocations only within that local framework. Consequently, events classified here as insertions may in some cases represent translocations or duplications originating from other genomic regions.

Assessing clustering consistency with phylogeny and block structure

To validate our clustering approach, we first assessed whether the resulting clusters are compatible with the core genome phylogeny. For each junction, cluster assignments were mapped onto the core genome phylogenetic tree. We then applied the Fitch parsimony algorithm to determine the minimum number of evolutionary events required to explain the observed cluster assignments while allowing for one background cluster that is assumed to have been present at the root. We identified clusters that cannot be explained by a single parsimonious event, which we expect to occur rarely given the short evolutionary timescale considered in this study.

We further evaluated how well the clustering reflects the observed block structure, both at the block level and at the edge level. To do this, we computed the Jaccard index [96] based on the presence or absence of blocks or edges between each pair of isolates:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (3.1)$$

Here, A and B denote the sets of blocks or edges present in isolates A and B , respectively. To assess the quality of the clustering, we compared within-cluster and between-cluster similarities. Specifically, we used the silhouette score [97], defined as

$$s(i) = \frac{\delta_b(i) - \delta_w(i)}{\max(\delta_b(i), \delta_w(i))} \quad (3.2)$$

where $\delta_w(i)$ is the mean distance of sample i to other samples within the same

cluster and $\delta_b(i)$ is the mean distance to samples in the nearest neighboring cluster. For each sample, we computed the mean within-cluster and between-cluster Jaccard indices. Since the silhouette score is defined on distances, the Jaccard index was converted to a distance measure using $d_J = 1 - J$. To obtain a silhouette score for each junction, the silhouette scores of all isolates belonging to that junction were averaged.

Identifying external origins of inserted sequences

To investigate the potential origins of inserted sequences beyond the genomes in which they were detected, we searched them against the AllTheBacteria reference database [62], which contains approximately 2.4 million assembled bacterial genomes. To efficiently query this large collection, we used LexicMap [61], a nucleotide alignment tool designed for searching moderate-length sequences against millions of prokaryotic genomes. LexicMap indexes the database using a small set of probe k-mers and stores the resulting seeds in a hierarchical index, enabling fast and memory-efficient sequence alignment at this scale.

Table 3.1: LexicMap search parameters used for different query length regimes. Alignment constraints include minimum percent identity, coverage thresholds, and minimum alignment length. Seed parameters control the minimum prefix length required for seed matches during the search.

Query Length	Alignment Constraints				Seed Parameters	
	pident	qcov_hsp	qcov_genome	align_len	-p	-P
Short (200 – 1000 bp)	98	90	90	200	15	17
Medium (1000 – 2500 bp)	98	70	90	1000	17	19
Large (> 2500 bp)	98	50	80	1000	19	21

Because our goal was to identify the likely origins of inserted sequences, we restricted the search to highly similar matches with at least 98% sequence identity using `-align-min-match-pident 98`. The parameters `-min-qcov-per-hsp` and `-min-qcov-per-genome` control the fraction of the query that must be covered by a single high-scoring pair (HSP) or by all HSPs within a genome, respectively, and `-align-min-match-len` restricts the minimal length required for the highest scoring pair. For short queries, we expect near-complete coverage by a single alignment, whereas longer queries may be split across multiple alignments due to fragmented assemblies or circular plasmids represented linearly in the reference. To account for this, we allowed lower single-HSP coverage thresholds for longer queries. The seed parameters `-p` and `-P` define the minimum prefix length required for seed matches

and thus balance search sensitivity and computational efficiency. All used parameters are summarized in table 3.1. Queries shorter than 200 bp were excluded, as they could be artefacts of pangenome graph construction. We restrict our analysis to 10000 hits per query to avoid losing high quality hits due to LexicMap k-mer prefiltering. For each hit, Lexicmap reports the corresponding BioSample accession number, which enables retrieval of additional metadata from NCBI [98], including organism, strain, serovar, and sequence type.

Chapter 4

Results

Homologous recombination shapes junction clustering but only rarely introduces structural variation.

Homologous recombination enables the integration of foreign genetic material into bacterial genomes by requiring two largely homologous sequences flanking the integration site. This process can either introduce novel cargo genes or result in the deletion of pre-existing genomic regions. Previous studies have shown that such events frequently occur between conserved core genes flanking hotspot junctions [20], making their detection a necessary prerequisite for identifying downstream structural variations. Homologous recombination often leaves signatures of elevated sequence divergence in the host genome [21]. To identify these, we analyzed core genome blocks flanking each junction by constructing phylogenetic trees from their alignments and detecting branches exceeding a defined divergence cutoff. The cutoff of 0.001 substitutions per site was chosen based on the cumulative distribution of branch lengths across all trees (Supplementary Fig. 1). This distribution shows a separation between a large fraction of short branches (on the order of 10^{-5} – 10^{-4}), likely reflecting background mutations, and a smaller group of longer branches (> 0.001), indicative of recombination-driven divergence. The combined length of flanking core block alignments consistently exceeded 1000 bp (Supplementary Fig. 2), ensuring that single random mutations do not inflate branch lengths beyond 0.001 substitutions per site and thus minimizing false-positive detections.

We further validated the chosen cutoff by assessing how well the resulting clusters

Table 4.1: Structural event statistics and clustering agreement across branch-length thresholds. For each branch-length cutoff used to cluster junctions based on sequence similarity in flanking core blocks, the table reports clustering summary statistics, clustering consistency statistics with the general phylogeny, counts of structural event types, and median silhouette scores computed from block- and edge-based Jaccard similarity. Increasingly conservative cutoffs reduce the number of clusters and clustered junctions, while phylogenetic inconsistency, the number of inferred structural events, and the silhouette scores remain largely stable across thresholds. Consistent with the branch-length distribution observed elsewhere, a cutoff of 0.0005 appears too permissive to reliably distinguish elevated sequence diversity caused by homologous recombination from variation due to random mutation. Accordingly, the clustering results begin to show slight distortions at this threshold.

Cutoff	Clustering Summary		Phylogenetic Consistency	
	Clusters in clustered junctions	Clust. junctions	Inconsist. clusters	Inconsist. junctions
0.0005	731	247	13	13
0.001	562	210	10	10
0.002	485	177	11	11
0.003	414	148	12	12
0.004	367	131	10	10
0.005	394	106	8	8
0.006	234	83	10	10

Cutoff	Structural Events				Silhouette		
	Insertions	Deletions	Translocations	Inversions	Total Events	Block JI	Edge JI
0.0005	1778	285	23	25	2111	-0.10	-0.14
0.001	1765	287	49	25	2126	0.00	0.00
0.002	1780	291	52	26	2149	0.00	0.00
0.003	1801	306	56	27	2190	-0.04	-0.01
0.004	1814	302	55	25	2196	-0.04	-0.02
0.005	1806	302	51	25	2184	0.00	0.00
0.006	1811	308	51	25	2195	0.00	0.00

align with the *E. coli* ST131 core genome phylogeny. For cutoffs below 0.001, the number of phylogenetically inconsistent clusters increases, whereas it remains largely stable across cutoffs from 0.001 to 0.006 (Table 4.1). At the selected cutoff of 0.001, we identify 10 phylogenetically inconsistent clusters across 10 junctions (0.5% of clusters, 1.9% of junctions). All occur within the majority cluster and can each be explained by two independent parsimonious events. In five junctions, the isolate *NZ_JAPVBX010000001.1* clusters with clade C rather than with clade B, where it is placed in the core genome phylogeny. This can be explained by an additional homologous recombination event involving material from clade C. In four of these five cases, this event is accompanied by a structural exchange of accessory material, yielding two longer and two shorter paths. In the remaining five junctions, the majority cluster of clade C also appears as a subcluster within clade A, representing only a minor inconsistency with the phylogeny. As expected, increasing the

cutoff leads to a reduction in both the number of clusters and clustered junctions (Table 4.1).

To further evaluate cluster quality, we assessed how well clustering reflects block structure diversity at each junction. Silhouette scores based on Jaccard indices, computed from the presence or absence of shared blocks or edges between isolate paths, quantify separation between and within clusters. Across cutoffs, median silhouette scores remain close to zero and deteriorate for cutoffs below 0.001, indicating limited cluster separation. Notably, the distribution of silhouette scores is bimodal, with clusters showing either high or low values (Supplementary Fig. 3a). This pattern suggests that while some clusters capture genuine structural differences, others separate isolates with identical block structures, consistent with homologous recombination events that do not alter genomic content. At the selected cutoff of 0.001, restricting the analysis to more complex junctions further supports this interpretation (Supplementary Fig. 3b), as singleton and binary junctions, typically involving few structural changes, exhibit particularly low silhouette scores. In these cases, downstream structural events appear too infrequent to provide a sufficiently strong clustering signal. Because silhouette scores are strongly affected by split clusters, they may not fully capture clustering quality in this setting. We therefore additionally assessed whether clustering agrees with changes in block structure by examining binary, non-singleton junctions. In particular, we examined how often structural events were split across multiple clusters and thus could not be explained by a single evolutionary event. This was observed in only 2 of 49 binary, non-singleton junctions, indicating that such cases are rare and supporting the overall validity of the clustering approach. In the junction *VRDEBAMMSO_r_WXCHSHHCDT_r*, two clusters are observed, one of which contains an additional block in all but two isolates. This pattern can be explained either by an insertion affecting the full cluster followed by a deletion in two isolates, or by two independent deletion events. In the junction *RYJVCTXEHC_r_STVEZQVFDI_r*, two independent single insertion events with largely identical content occur in two separate clusters.

For SNPs within flanking core blocks, we also examined whether sequence differences were concentrated within a single core block or a specific subregion. Instead, they appeared to be distributed relatively uniformly across the combined flanking core blocks. However, this observation should be investigated more systematically in future work. Finally, we assessed the sensitivity of inferred downstream structural event counts to the choice of cutoff. Across the selected range of moderate cutoffs, event counts remained largely stable (Table 4.1). Overall, the final cutoff of

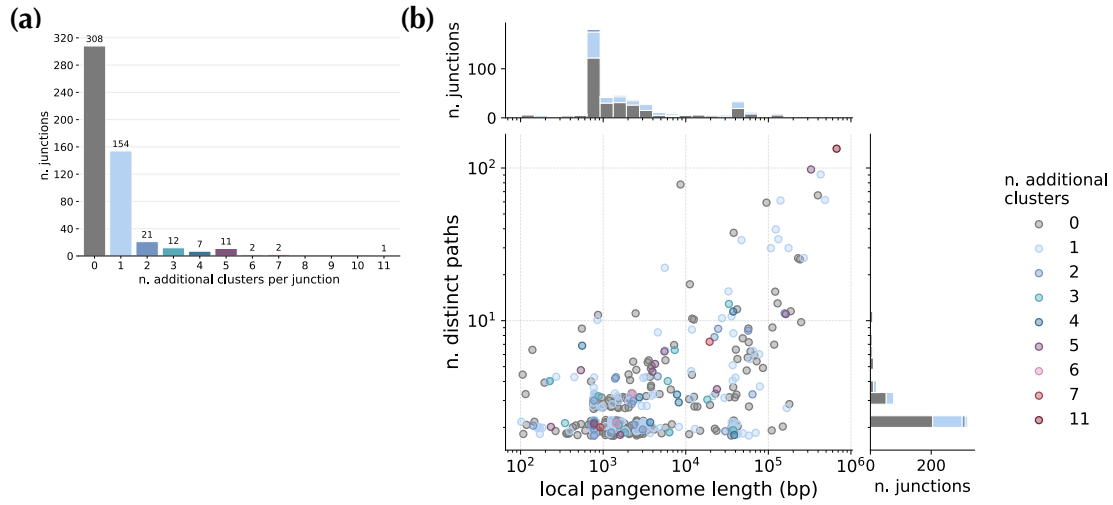


Figure 4.1: Additional clusters are frequently observed across junctions but show limited association with junction diversity. (a) Frequency distribution of additional clusters per junction, indicating that additional clusters are present in slightly less than half of all junctions, whereas junctions containing multiple additional clusters are comparatively rare. (b) Association between the number of additional clusters and junction diversity, quantified by local pangenome length and the number of distinct paths. Points are colored by cluster count. No clear association between cluster abundance and junction diversity is observed.

0.001 appears to provide a robust separation between homologous recombination and background mutation, while remaining stable to moderate increases in the threshold.

After defining the branch-length cutoff, we used it to assess the extent to which junctions are affected by homologous recombination. Of the 518 junctions considered, 210 (40.5%) were subdivided into clusters based on sequence divergence in the flanking core blocks. Among these, 154 junctions (73.3%) contained exactly one additional cluster. Higher numbers of additional clusters were progressively less common, with a maximum of 11 observed for a single junction (Figure 4.1a). In total, we observe 352 additional clusters, approximately one every 20 core genome substitutions. We next examined whether the number of additional clusters was associated with junction diversity, measured by combined accessory genome length and the number of distinct paths [21]. No clear relationship was observed between cluster abundance and junction complexity (Figure 4.1b).

To investigate how often structural changes occur together with, or shortly after, homologous recombination, we focused on binary junctions. These can be explained by a single gain, a single loss, or a combination of both, and therefore allow a comparatively straightforward classification of whether a structural event is associated with homologous recombination. We distinguished events present in all isolates of

a single cluster from those found only in a subset of a cluster, which likely arose downstream. Among singleton junctions, only 6 of 273 events (2%) were associated with homologous recombination. In contrast, among non-singleton junctions, 8 of 49 events (16%) were confined to a single cluster. One of these corresponded to a combined gain-and-loss event, a category observed three times among non-singleton junctions. Table 4.2 summarizes the results for singleton junctions, where gain and loss events can be assigned with high confidence based on their frequency distribution. Overall, some of these events showed signatures suggestive of an association with mobile genetic elements, whereas others did not.

Table 4.2: Classification of singleton junctions into gain and loss events and their association with homologous recombination. Singleton junctions are categorized into those consistent with a single gain event, a single loss event, or a combination of both. For each category, we assess whether clustering of isolates yields a cluster containing only the affected isolate, which would suggest that the event is associated with homologous recombination.

	n. junctions				n. associated with homologous recombination			
	Gain	Loss	Gain + Loss	Total	Gain	Loss	Gain + Loss	Total
Singleton	245	17	11	273	3	1	2	6

Lastly, we examined the length of structural changes associated with homologous recombination. Because assigning individual clusters to the recombination event that generated them becomes increasingly ambiguous in junctions with many clusters, we restricted this analysis to junctions with exactly one additional cluster. In the majority of these junctions, the clustering cleanly separated clade A from clades B and C (see Figure 1.2 for the overall phylogeny). Of the 154 junctions considered, 130 (84.4%) showed no associated structural change. In the remaining 24 junctions, the median length of the affected sequence was 2565 bp (IQR: 789–12454 bp), with a distribution skewed toward shorter events. Taken together, these findings indicate that homologous recombination frequently occurs without detectable structural consequences, whereas a minority of cases are accompanied by changes in genome structure that must be accounted for in downstream event detection, underscoring the necessity of clustering.

Mobile genetic elements are the main driver of accessory genome diversity

To detect structural events occurring downstream of homologous recombination, we first defined a reference path for each cluster as a consensus path representing the

block structure of the most recent common ancestor along the core genome phylogeny. Because 13.6% of blocks are duplicated, we first deduplicated blocks using context anchors. In cases where repeated duplications or non-informative intervening blocks prevented unambiguous assignment, blocks were distinguished by their genomic order. Although this can introduce errors in isolates where only one copy is retained, such cases were rare. After global deduplication across all isolates and clusters, 2.5% of blocks remained ambiguous and required numbering, corresponding to 18.4% of duplicated blocks. Local deduplication within each cluster, combined with the exclusion of isolates with too few informative context anchors, further reduced this fraction to 2.1%.

Block presence or absence in the consensus path was inferred from the reconstructed root state. In 3.4% of blocks, spanning 21 junctions (4.1% of all junctions), the Fitch algorithm returned an ambiguous root state. For blocks observed at least twice, we resolved this ambiguity using within-block sequence similarity. Among ambiguous blocks, 29.7% occurred only once and were classified as gains, 24.0% showed low sequence similarity and were interpreted as independent gains, and 46.3% showed high sequence similarity and were classified as losses.

Structural events, including insertions, deletions, translocations, and inversions, were then detected relative to the consensus reference. At this stage, context no longer needed to be defined uniquely across all isolates, but could instead be assigned separately for each isolate–consensus pair. This avoids cases in which duplications or inversions in other isolates obscure the relevant local context. This is particularly important for translocation detection, which requires context anchors for all blocks. Nevertheless, 2.1% of blocks remained in ambiguous contexts and therefore still required numbering. To avoid overcounting events observed in multiple isolates, detected events were subsequently deduplicated based on unique block structure and global context, using the previous global context definition, in which 2.5% of blocks remained ambiguous after deduplication.

Translocations are identified as blocks occurring in different genomic contexts. Ambiguity arises when a translocated block was also deleted once and therefore appears at multiple positions in the consensus path, or when two translocation events with identical block structure are observed. This occurred only once, corresponding to 2% of all translocations. In addition, translocations cannot always be distinguished from separate deletion and insertion events. Insertions can similarly be ambiguous when they occur within duplicated blocks lacking unique context anchors. In such

cases, the affected copy cannot be distinguished if only one copy was inserted. This does not usually change the number of inferred insertions, but rather their assigned location. Such cases accounted for 5.6% of deduplicated insertions (99 events across 25 junctions). Incorporating sequence similarity could help resolve these ambiguities in future work. Altogether, although heuristic rules were required to resolve a small number of ambiguous edge cases, these were rare and unlikely to substantially affect event counts or the inferred locations of most events.

In total, we detected 2126 structural events, corresponding to approximately one structural event every 3 core genome mutations. Most events were insertions (1765; 83.0%), followed by 287 deletions (13.5%), 49 translocations (2.3%), and 25 inversions (1.2%). Accordingly, gain events occur approximately every 3 core genome mutations, whereas loss events occur every 20 mutations on average, translocations and inversions are observed approximately every 125 and 250 mutations, respectively. These rates are estimated using the total tree length as a reference and should therefore be regarded as approximations.

Structural events were then annotated by their association with mobile genetic elements. Insertions, translocations, and inversions were classified as insertion sequences or prophages if the annotation and structural event overlapped by at least 80%. Events were classified as associated with a defense system, integron, insertion sequence, or prophage if at least 80% of the annotation was contained within the structural event. When multiple annotations overlapped the same event, labels were assigned in the priority order integron > defense system > prophage > insertion sequence. Overall, 86.6% of events were classified as, or associated with, mobile genetic elements, suggesting that these are the major drivers of structural diversity downstream of homologous recombination. Among annotated events, 410 (25.8%) were associated with more than one element. A majority of annotated events were identified as insertion sequences (989; 62.1%), followed by prophages (301; 18.9%). In addition, 15 events were associated with integrons (0.9%), 165 with defense systems (10.4%), 39 with prophages (2.4%), and 438 with insertion sequences (27.5%). More generally, only 34.1% of all annotations overlapped structural events. These annotations may represent mobile genetic elements introduced by homologous recombination or inserted before the inferred structural event. However, because this estimate was obtained without deduplicating annotations, it should be regarded as a broad approximation.

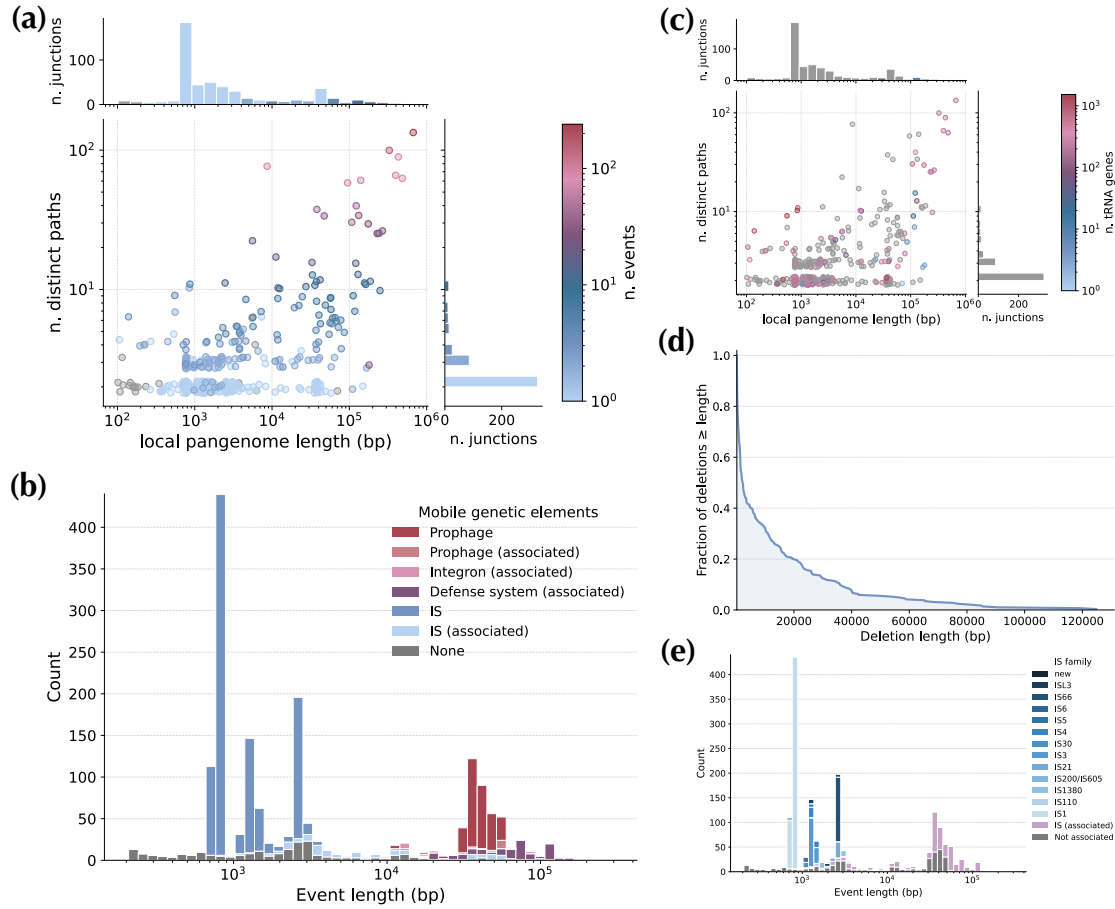


Figure 4.2: Structural event diversity, length, and mobile genetic element associations across junctions. (a) Relationship between the number of structural events detected per junction after clustering, including insertions, deletions, inversions, and translocations, and junction diversity, quantified by local pangenome length and the number of distinct paths. Most detected events are insertions. Points are colored by the number of events, and marginal bars indicate the distribution of junctions colored by the median number of events per bar. Structural event counts increase with junction diversity. (b) Length distribution of insertions colored by mobile genetic element annotation. A prominent insertion-sequence peak is observed between 700 bp and 4000 bp, and a prophage-associated peak between 20 and 50 kbp. Longer insertions are predominantly associated with defense systems. (c) Relationship between the number of tRNA genes observed in a junction and junction diversity. tRNA genes are found across the full range of junction complexity, with a slight enrichment in more complex junctions. (d) Reverse cumulative distribution of deletion lengths. Most deletions are short, whereas longer deletions are rare. Deletion lengths range from 200 bp to 125 000 bp. (e) Length distribution of insertions colored by their association to insertion sequence family members. Distinct peaks are associated with different families, including a major IS1 peak around 700–900 bp, an IS3 peak around 1000–1100 bp, and an IS66 peak around 2500 bp. Notably, many insertions in the prophage size range are also associated with insertion sequences, suggesting secondary insertion into prophage-associated regions.

Structural event counts per junction were then examined in relation to junction diversity, quantified by the number of distinct paths and the combined accessory genome length. In contrast to the number of clusters, event counts were strongly

associated with junction diversity (Figure 4.2a). The same pattern was observed for each event type separately (Supplementary Fig. 4), suggesting that structural events, many of which are likely driven by mobile genetic elements, are a major source of junction diversity. To further support this interpretation, we assessed whether junctions containing tRNA genes show increased diversity (Figure 4.2c), as mobile genetic elements using integrases often integrate adjacent to tRNA genes [22]. tRNA-associated junctions were observed across the full range of junction diversity, with a slight enrichment among more complex junctions. This indicates that accessory genome junctions are frequently associated with tRNA genes and suggests a modest trend toward higher complexity in junctions shaped by repeated integration of mobile genetic elements.

We next examined the length distributions of different event types. For deletions, inversions, and translocations, short events were common whereas longer events were progressively rarer. Deletions ranged from 200 bp to 125 000 bp, inversions from 200 bp to 65 000 bp, and translocations from 300 bp to 35 000 bp. The reverse cumulative distribution of deletion lengths is shown in Figure 4.2d. Inversions and translocations followed similar patterns, although both were represented by relatively few events. Insertion lengths, in contrast, showed distinct peaks corresponding to different classes of mobile genetic elements (Figure 4.2b). We observed a prominent insertion-sequence peak between 700 and 4000 bp and a prophage peak between 30 and 50 kbp. Longer insertions were predominantly associated with defense systems, which may reflect integrative conjugative or mobilizable elements (ICEs/IMEs), although not exclusively. Examining the insertion-sequence peak in more detail, we further annotated insertion sequences by family (Figure 4.2e). Distinct peaks were associated with different families, including a major IS1 peak around 700–900 bp, an IS3 peak around 1000–1100 bp, and an IS66 peak around 2500 bp. Among 973 insertion-sequence insertions, 528 belonged to the IS1 family, 134 to IS3, and 135 to IS66. We also note numerous insertion-sequence-associated elements within the prophage size range, suggesting secondary insertion of insertion sequences into prophage-associated regions.

Genomic origins of detected insertions

The next objective of our analysis was to investigate the origin of the detected insertions. We first searched for potential origins within the host isolate itself, including both the chromosome and its plasmids. Each insertion was queried individually, re-

quiring at least 99% sequence identity and 90% query coverage while accounting for linearly represented chromosomes. Identical insertions were then deduplicated by summarizing all occurrences using the median number of hits across occurrences. Of the queried insertions 959 (54.3%) showed additional hits in the host chromosome, 539 (30.5%) were found only on plasmids, and 267 (15.1%) were not detected in either compartment, suggesting a potential external origin. In total, 767 insertions (43.5%) were found in both chromosome and plasmids. Figure 4.3a–b summarizes the distribution of additional hits in the host chromosome and plasmids, as well as the combined number of hits across both compartments. Most insertions were detected up to 10 times, with progressively fewer insertions at higher copy numbers; the maximum combined number of hits observed was 43.

Because not all insertions could be identified within the host chromosome or plasmids, we next searched for potential external origins in the AllTheBacteria database, using a maximum cutoff of 10 000 hits. We retained only hits with 100% identity and at least 90% coverage. As before, identical insertions were deduplicated by summarizing repeated occurrences with the median number of hits. For 690 insertions (39.1%), no external matches were found. Among the insertions with external matches, we observed both a peak of rare insertions and a substantial number with more than 5,000 hits (Figure 4.3c). The latter likely represents only a fraction of the true number of matches, as the 10,000-hit limit and approximate seed finding in LexicMap may underestimate highly abundant sequences. In many cases, the number of total hits exceeded the number of genomes with hits, indicating that these insertions are often duplicated within genomes.

For all retrieved external hits, we queried BioSample identifiers in NCBI to assign species labels. Only 52 of 448,166 unique identifiers could not be resolved. As expected, most hits were assigned to *E. coli* (3,406,411), followed by *Salmonella enterica* (424,212), *Shigella flexneri* (212,633), and *Klebsiella pneumoniae* (200,278). *Shigella sonnei* was less frequent (21,901 hits), while all other species occurred fewer than 10,000 times, with approximately 270 species represented overall. Notably, 44,172 of the *E. coli* hits were identified as belonging to the ST131 strain (1.3 %). To examine how species composition varied across insertion frequencies, we colored the distribution of the number of external genomes by the relative abundance of the five most frequent species (Figure 4.3d). This revealed that *Klebsiella pneumoniae* was more often associated with low-frequency hits, whereas *Salmonella enterica* and *Shigella flexneri* were more common among insertions with high numbers of external matches. When considering only the majority species assigned to each insertion,

we found 1,027 insertions dominated by *E. coli*, 46 by *Salmonella enterica*, 11 by *Klebsiella pneumoniae*, two by *Shigella flexneri*, one by *Shigella sonnei*, and one by *Acinetobacter baumannii*. This distribution is shown in Supplementary Figure 5a. Stratification by insertion length (Supplementary Figure 5b) indicated that external sequence origins were observed primarily for shorter insertions within the insertion-sequence peak.

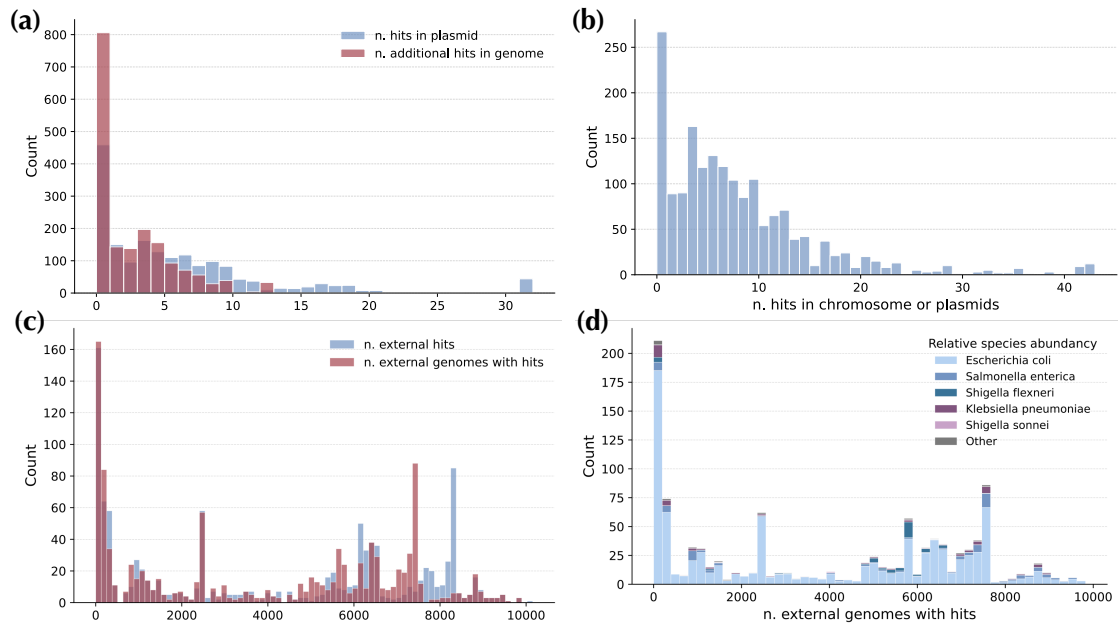


Figure 4.3: Potential origins of detected insertions within the host genome, plasmids, and external genomes. We first examined whether detected insertions had additional matching sequences within their host isolate, either in the chromosome or on plasmids. Panel (a) shows the distribution of additional hits in the host chromosome and on host plasmids, while panel (b) shows the distribution of the combined number of hits across both compartments. We identified 806 insertions with no additional hits in the host chromosome and 459 with no additional hits on host plasmids. In total, 267 insertions were not found in either compartment. Overall, when additional copies were present, insertions were most commonly detected between 1 and 10 times, with decreasing abundance at higher copy numbers. The maximum combined number of hits observed was 43. We next investigated the occurrence of the identified insertions in the AllTheBacteria database, using a maximum cutoff of 10,000 genome hits. Panel (c) shows the distribution of total hits and the number of external genomes with matches at 100% identity and at least 90% coverage, excluding insertions with no external matches. We observed both a peak of rare insertions and a substantial number of insertions that were detected in more than 5,000 genomes. Finally, panel (d) summarizes the species distribution of external matches, restricted to the five most abundant species. For each range of external genome hits, the relative species abundance is shown. As expected, most matches were found in *Escherichia coli* isolates, followed by *Salmonella enterica*, *Shigella flexneri*, and *Klebsiella pneumoniae*, whereas *Shigella sonnei* was less frequent. External species annotations were observed for both low and high insertion-hit ranges.

Overall, many insertions appear to originate from elsewhere in the host genome or from host plasmids. When no internal origin was detected, the most likely source

was other isolates of the same species. Matches to other species were comparatively rare and usually found in closely related taxa.

Interactive visualization of accessory genome junctions

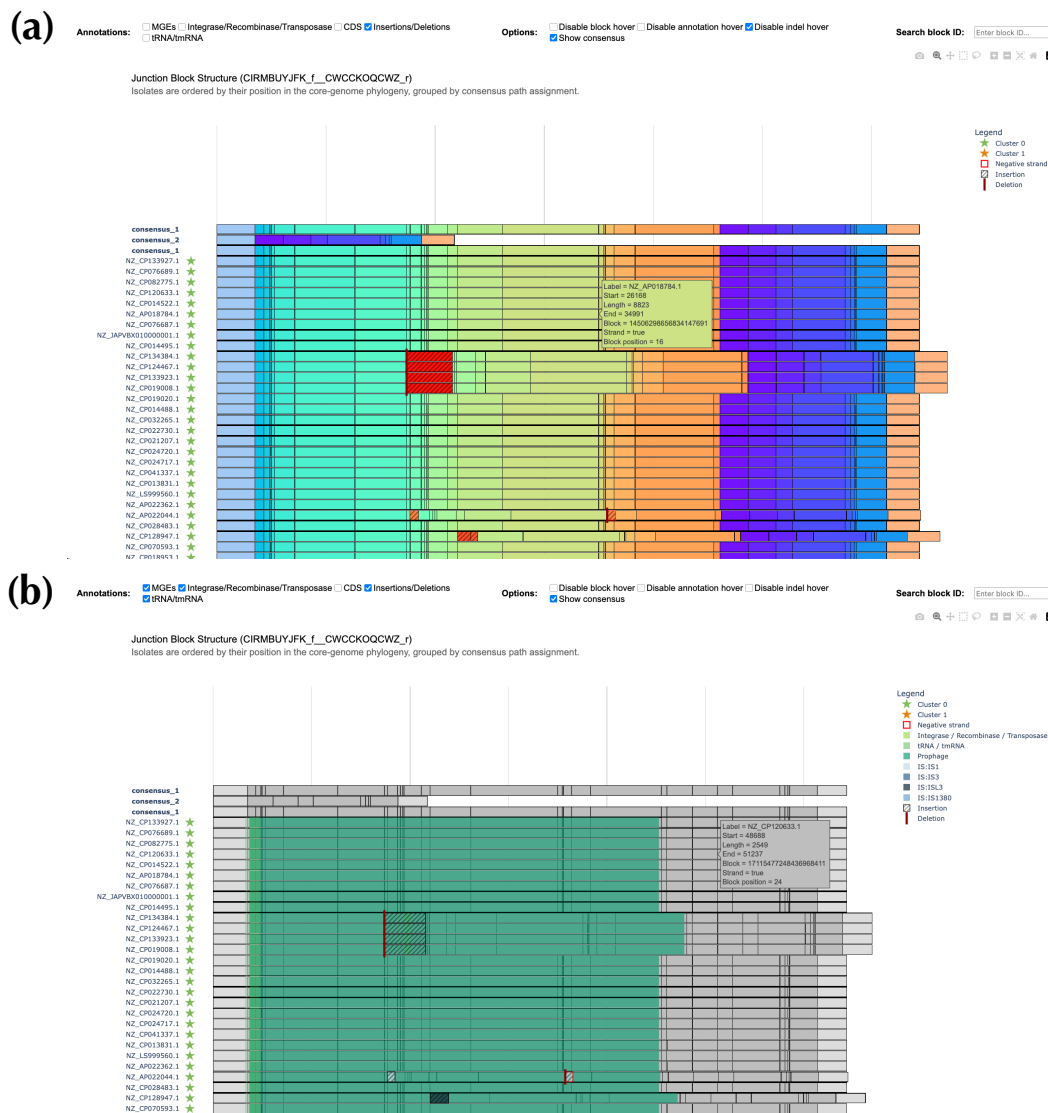


Figure 4.4: Interactive dashboard for exploring accessory genome junctions and associated genomic annotations. General layout of the dashboard for the exploration of accessory genome junctions. It contains a colorful mode that highlights block structure (a) and a grey mode used to show subsequent annotations (b). Different checkboxes can be used to enable or disable annotation categories. Structural events can be visualized in both modes.

To enable interactive exploration of accessory genome junctions, we developed a web-based visualization dashboard using **Dash** (version 3.4.0) and **Plotly** (version 6.0.0) [99]. The dashboard brings together the results obtained in the previous analyses, including junction clustering, consensus paths, inferred evolutionary events, and functional annotations, in a single interface (Figure 3.1d, 4.4). For a selected junction, the dashboard displays the full pangraph as a horizontal block diagram. Isolates are ordered according to the core genome phylogeny and grouped by their consensus cluster, although clustering can be disabled to show isolates solely in phylogenetic order (Supplementary Fig. 7b). Each genomic block is represented as a colored bar, with core and accessory blocks distinguished by shading. Consensus paths are shown as additional aligned rows, and colored stars indicate the assigned clusters. If a secondary clustering is provided, the stars can additionally be used to compare alternative clusterings and highlight potential inconsistencies. Different annotation layers can be toggled interactively using checkboxes. These include mobile genetic elements associated annotations such as insertion sequences, prophages, integrons and defense systems, as well as integrase, recombinase, and transposase annotations, coding sequences, tRNA and tmRNA genes, and detected structural events including insertions, deletions, inversions, and translocations (Supplementary Fig. 6). A search function allows users to highlight all occurrences of a given block ID across isolates (Supplementary Fig. 7a). To improve readability in dense regions, hover interactions can be enabled or disabled for individual annotation layers. The dashboard is available on GitHub.

By combining structural diversity, phylogenetic relationships, and functional annotations of mobile genetic elements and gene content, the dashboard provides an intuitive overview of the evolutionary history encoded in each junction. This is particularly valuable for hotspot junctions, whose complexity is often difficult to interpret from static representations alone (Supplementary Fig. 8). The dashboard proved highly useful throughout this study for exploring and understanding the patterns described in the analyses above.

Chapter 5

Discussion

In this study, we combined local sequence similarity, structural information, and isolate relationships derived from the core genome phylogeny to disentangle diversity arising from homologous recombination from secondary structural variation. This approach enabled us to reconstruct the evolutionary processes shaping the accessory genome in a dataset of 222 closely related *E. coli* ST131 isolates across a short evolutionary timescale. Focusing on high-complexity junctions, we characterized their genetic content and explored the likely origins of inserted sequences. Overall, our results show that accessory genome junctions in ST131 are shaped primarily by repeated activity of mobile genetic elements (MGEs), while homologous recombination contributes importantly to sequence divergence in the flanking core blocks and to clustering structure, but only rarely introduces major structural changes itself.

A central step in this study was the identification of clusters within accessory genome junctions. This was necessary because recombination events affecting the flanking core genome can create groups of isolates with substantially different sequence backgrounds [20], potentially obscuring the interpretation of accessory genome changes if all isolates are analyzed together. Clustering based on sequence divergence in the flanking core blocks was therefore biologically motivated. When evaluated in the context of block structure and species relationships from the global core genome phylogeny, the resulting clusters were broadly consistent and inferred downstream results remained robust to moderate changes in the clustering cutoff of 0.001 substitutions per site, increasing our confidence in the cluster definition.

We also explored whether clustering could be based on structural information alone. To this end, we filtered out rare blocks and isolate paths with rare edges to reduce

the influence of downstream events, selected the remaining paths as cluster seeds, and assigned isolates according to edge Jaccard similarity. In practice, however, this approach proved unreliable because cluster sizes varied substantially, making it difficult to define rarity in a consistent way. Depending on the threshold, structural clustering either artificially split clusters or failed to recover necessary ones. Sequence-based clustering was therefore preferable, as it produced biologically interpretable groups and better separated changes associated with recombination from those that occurred later along individual branches.

Although, our clustering is a consequence of homologous recombination, the number of observed clusters cannot be interpreted directly as the number of underlying recombination events. There are several reasons for this. First, clusters are defined by cutting internal branches whose length exceeds the chosen cutoff. Depending on the topology of the tree, however, a single cut does not necessarily create a new cluster. In addition, homologous recombination may have occurred multiple times along the same branch, although we expect this to be relatively rare on the short timescales considered here. Another possible, though likewise rare, scenario is that two independent homologous recombination events integrate the same material, which would cause us to merge two unrelated clusters into one. In such cases, incompatibilities with the core-genome phylogeny can arise, which we observed only rarely. From a biological perspective, homologous recombination is most likely to occur between identical or highly similar sequences [65]. Such events would probably not be detected by our algorithm, but they are also less likely to produce structural changes that obscure the identification of downstream events, and they did not appear to be problematic in this study. In addition, recombination events may occur exclusively within the accessory genome, leaving no signal in the flanking core-genome blocks. Finally, our approach may overcount homologous recombination events because we treat junctions only in their local context. In reality, core-genome blocks between junctions can overlap, and a single homologous recombination event may span multiple junctions. For all of these reasons, clustering should be interpreted as a signature that homologous recombination has occurred, rather than as a precise measure of the number or rate of events. Future work could aim to identify homologous recombination regions more systematically across the entire core genome, or even within the accessory genome, to provide a clearer picture.

Our clustering results suggest that a large fraction of the observed junction diversity arises independently of homologous recombination. More than half of all junctions, spanning a broad range of diversity levels (Figure 4.1b), were not subdivided into

additional clusters. Where clustering was observed, it was most often limited to a single additional cluster, frequently separating clade A from clades B and C in agreement with the known population structure of ST131 [51], whereas higher numbers of clusters were uncommon. Together, these observations support the choice of a short evolutionary timescale, in which homologous recombination contributes to variation but does not dominate the overall signal. Considering within- versus between-cluster structural variation in blocks and edges, the lack of a clear relationship between cluster number and junction complexity, the distribution of events in binary junctions, and accessory genome differences between clustered groups, we conclude that homologous recombination usually does not introduce major structural changes. This is consistent with the general deletion bias observed in bacterial recombination, which tends to favor the integration of shorter DNA segments due to DNA degradation [6]. Nevertheless, we also identified rare cases in which structural changes appeared to be associated with homologous recombination despite the absence of clear MGE signatures. Given that *E. coli* is generally not considered naturally transformable, such DNA may instead have been introduced by generalized or lateral transduction [74] or by other non-canonical transfer mechanisms [20]. The limited role of transformation in *E. coli* suggests that homologous recombination may play a more prominent role in shaping genome structure in naturally transformable species. Interestingly, Oliveira et al. reported an enrichment of homologous recombination within hotspot regions across multiple bacterial species, including *E. coli* [20], a pattern that we did not observe in our ST131 dataset.

Before interpreting downstream structural events in detail, it is important to acknowledge the limitations of the consensus path reconstruction and event detection framework. Both steps rely on heuristic decisions and are therefore not expected to resolve every junction perfectly. In particular, ancestral block presence and absence cannot always be inferred unambiguously from the tree. Although block frequency and sequence similarity provide useful criteria for resolving such cases, they represent only the most likely explanation and may occasionally be incorrect. Additional ambiguities arise during block deduplication, especially when duplicated blocks lack informative context anchors that distinguish them. To address this, a small number of isolates had to be excluded from consensus definition in some junctions because too few reliable anchors were available, while still ensuring that enough isolates remained for meaningful consensus inference. Ambiguous deduplication can lead to the wrong duplicate of a block being identified as an insertion, although in most

cases this affects primarily the inferred location of the event rather than the overall insertion count. Translocations introduce further uncertainty, as multiple possible matching locations may exist in the consensus path, making their origin difficult to determine. In addition, they can often not be distinguished unambiguously from separate insertion and deletion events. Incorporating sequence similarity more explicitly into future versions of the framework may help to resolve some of these ambiguities. More generally, the current approach considers only the local context of a junction and therefore cannot determine whether an inferred insertion actually represents a translocation or duplication from another part of the genome, or whether a deleted segment may have reappeared elsewhere. The framework can also account for overlapping events only to a limited extent. Inversions are handled separately from insertions, deletions, and translocations, but other combinations of superimposed events remain difficult to resolve. This likely leads to undercounting of events in some cases, for example when insertions occur within pre-existing prophage or duplicated regions, or when previously acquired insertions are masked by subsequent larger deletions. A useful future extension would therefore be to add a second level of event detection that captures insertions occurring within larger inserted segments shared across multiple isolates. Nevertheless, these edge cases appeared to be rare, and the overall framework proved sufficiently stable to recover the main patterns of structural evolution. In this context, the interactive dashboard provides an important complement to the algorithmic analysis, as visual inspection often makes it easier to assess whether inferred events are biologically plausible, particularly in complex junctions with overlapping rearrangements.

The analysis of downstream structural events provides further insight into the dynamics of accessory genome evolution in ST131. In total, we identified 2126 structural events, corresponding to approximately one event per three core genome substitutions. This estimate should be regarded as a lower bound on the overall rate of structural change, since it includes only events inferred downstream of homologous recombination. Nevertheless, it is consistent with the estimate of at least 2000 structural modifications reported by Molari et al. [21]. The majority of detected events were insertions (1765; 83.0%), followed by deletions (287; 13.5%), translocations (49; 2.3%), and inversions (25; 1.2%). The strong excess of insertions on recent branches is likewise consistent with the findings of Molari et al., who focused on binary junctions and reported approximately ten times more gains than losses. In our analysis, which includes hotspot junctions, the excess is less pronounced, with roughly six times more insertions than deletions. This reduction is in line with their

observation that the gain bias is strongest in singleton junctions [21]. The predominance of gains raises the broader question of why bacterial genomes nevertheless remain relatively stable and compact in size [100]. One possible explanation is that deletion bias during homologous recombination subsequently removes excess DNA [6], thereby counteracting the accumulation of recent insertions. In addition, many insertions are likely to be deleterious, for example if they disrupt coding sequences or regulatory regions [75], and may therefore be purged by purifying selection [100]. Given the short evolutionary timescale considered here, and our specific focus on events downstream of homologous recombination, such cleanup processes may not yet have had sufficient time to act, resulting in an apparent excess of recent gain events.

Since clustering associated with homologous recombination explained only a small part of the observed structural diversity, we next assessed the contribution of downstream events. We find a strong correlation between the number of structural events and junction diversity (Figure 4.2a), indicating that local structural processes are a major driver of hotspot complexity. Notably, 87% of detected events were associated with mobile genetic elements, highlighting their central role in shaping accessory genome variation. This interpretation is further supported by a slight enrichment of tRNA genes in high-complexity junctions, consistent with their known function as preferred integration sites for integrase-mediated MGEs [22].

Interestingly, only about one third of MGE annotations were directly associated with inferred insertion events. The remaining annotations likely correspond to elements integrated prior to the homologous recombination events defining the clusters, or to elements introduced as part of larger recombination sequences. This highlights a limitation of our framework, which captures primarily recent structural events. Analysis of insertion length distributions (Figure 4.2b) further supports the dominant role of MGEs. We observe a clear peak corresponding to insertion sequences, a second peak consistent with prophage-sized insertions, and a tail of longer insertions, frequently associated with defense systems, likely representing integrative conjugative or mobilizable elements (ICEs/IMEs) [101].

Examining insertion sequences in more detail, we find that IS1 family elements dominate recent structural events, followed by IS3 and IS66. This differs from the genome-wide distribution reported by Molari et al., where IS3, IS66, and IS21 were more prevalent [21]. The relative enrichment of IS1 in our analysis suggests that certain IS families may be more active on short timescales, potentially reflecting

differences in transposition mechanisms, with only IS1 family members potentially using non-replicative transposition [102]. Finally, we observe instances of insertion sequences embedded within prophage regions, further emphasizing the importance of nested or secondary insertion events and motivating future extensions of the framework to capture such hierarchical structural variation.

Because our framework considers only the local context of each junction, it cannot directly reveal the external origins of inserted sequences. To address this, we compared detected insertions against the host genome, associated plasmids, and the large-scale AllTheBacteria database, which spans a broad range of prokaryotic genomes. While most insertions could be traced back to sequences within the chromosome or plasmids of *E. coli*, approximately 15% showed no matches in these sources, suggesting an external origin. We therefore searched for potential external matches in AllTheBacteria, restricting our analysis to exact matches with 100% sequence identity. For computational efficiency, the number of hits per query was limited to 10,000. However, due to the k-mer seed-based prefiltering used by LexicMap, some high-identity matches may still have been missed. In addition, database incompleteness and sampling bias represent further limitations of this analysis. Among the identified matches, the majority were still found in other *E. coli* genomes, indicating frequent exchange within the species. Beyond this, we identified insertions with closest matches in other members of the *Enterobacteriaceae*, including *Salmonella enterica* (46 cases), *Klebsiella pneumoniae* (11), *Shigella flexneri* (2), and *Shigella sonnei* (1) [103], as well as a single case linked to *Acinetobacter baumannii*. Notably, the *Shigella* species are now widely considered clones of *E. coli*, reflecting their close evolutionary relationship [10]. Horizontal gene transfer between different *Enterobacteriaceae* species is well documented [104], and the observed patterns are consistent with such interspecies exchange. In particular, both *Salmonella enterica* and *Klebsiella pneumoniae* are known reservoirs of extended-spectrum- β -lactamases [103], which also underlie antibiotic resistance in *E. coli* ST131 [50, 51]. This suggests that some of the detected insertions may reflect the exchange of antibiotic resistance determinants across species boundaries. The single insertion associated with *Acinetobacter baumannii*, a member of the *Moraxellaceae* family and an important multidrug-resistant pathogen in clinical settings [105], suggests that gene exchange may have occurred between these species and highlights the possibility of horizontal gene transfer across more distantly related taxa.

In addition to the limitations that were already discussed, it should be mentioned, that while other gene-based pangenome graph tools can interpret local indel variation disrupting coding sequences via frame shifts as "gene loss" [29, 106], PanGraph treats them like SNPs. As a result, these events cannot be explicitly analyzed within the present framework. Furthermore, our analysis relies on the accuracy of genome assemblies, and technical artifacts may in some cases be misinterpreted as structural variation. To mitigate this issue, we restricted our analysis to high-quality reference sequences. Finally, this study focuses exclusively on the *E. coli* ST131 lineage, which is strongly associated with clinical environments. As such, the observed patterns may not generalize to other *E. coli* strains or more distantly related species. In addition, extending these results to larger time-scales and correcting for heterogeneous rates of genome evolution across sites remains challenging.

Taken together, our results highlight the rapid diversification of the prokaryotic accessory genome and identify mobile genetic elements as the primary drivers of structural variation in *E. coli* ST131. While this study provides a detailed view of recent structural dynamics, several open questions remain. In particular, a more comprehensive quantification of homologous recombination across the full genome, and its interaction with accessory genome structure, would be essential to better understand the processes shaping these regions. It remains unclear to what extent recombination events affect entire junctions or only parts of them, and whether they can span multiple junctions. Methodologically, future work should aim to extend the current framework to capture nested structural events, improve the identification of translocations and insertions through the integration of sequence similarity, and incorporate phylogeny-aware deduplication. More accurate estimation of event rates, taking into account the temporal window following recombination, would further refine our understanding of the pace of accessory genome evolution. In addition, mapping recombination events along the phylogeny could clarify their role in mediating genomic gain and loss. More broadly, investigating the relationship between hotspot junctions driven by mobile genetic elements and those associated with frequent homologous recombination, as well as characterizing the gene content and insertion sequence composition of structural events, will be key to disentangling the underlying evolutionary forces. In particular, further study of interspecies gene exchange within the *Enterobacteriaceae* may provide important insights into the spread of adaptive traits such as antibiotic resistance. Finally, assessing whether homologous recombination plays a larger role in generating structural diversity in naturally transformable species represents an important direction for future work.

Together, these approaches will contribute to a more comprehensive understanding of the mechanisms governing accessory genome evolution.

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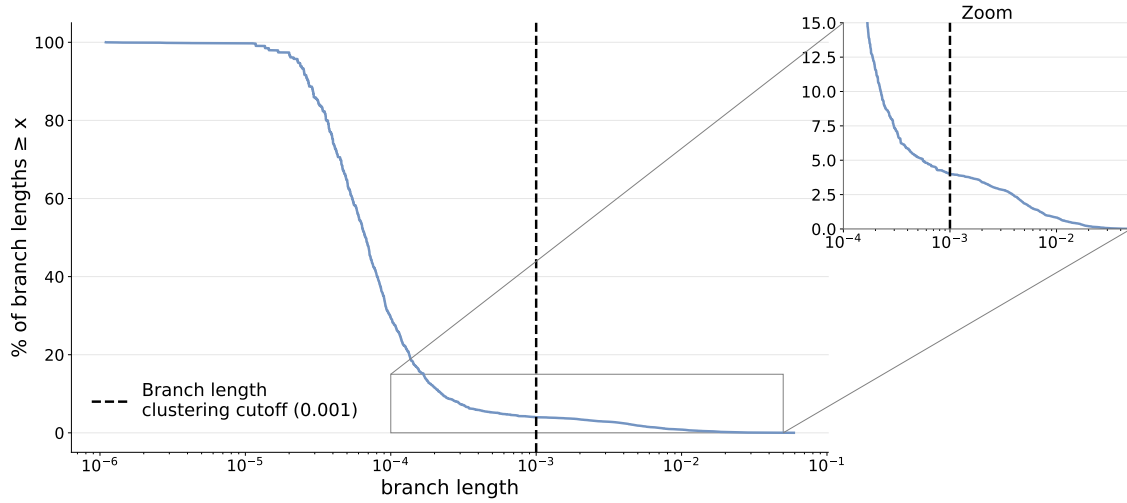


Figure 1: Complementary cumulative distribution of branch lengths in flanking core genome block trees. To distinguish sequence diversity arising from point mutations along the phylogeny from diversity introduced by homologous recombination, we determine a branch-length cutoff using the distribution of branch lengths in trees built from alignments of combined flanking core genome blocks. Most branches are shorter than 0.001, and the complementary cumulative distribution begins to flatten at this value. We therefore use 0.001 as a cutoff and interpret larger divergences as likely resulting from homologous recombination.

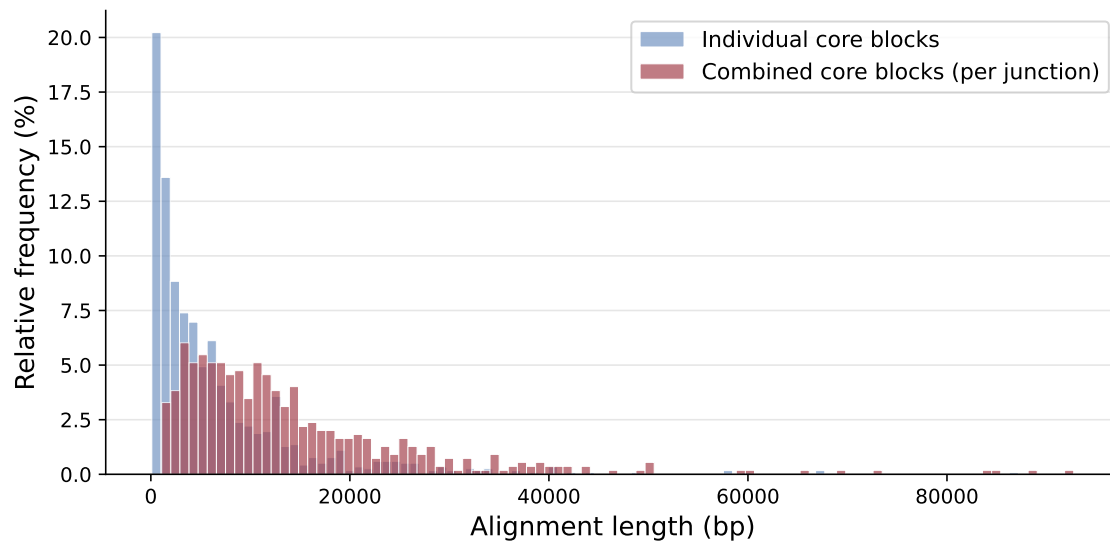


Figure 2: Distribution of core block alignment lengths. Junctions are clustered based on sequence similarity within their core blocks. With a clustering cutoff of 0.001, sequences differing by more than one base per thousand are separated. To prevent single mutations from affecting clustering, a minimum alignment length of 1000 bases is required. The figure shows the distribution of alignment lengths for individual core blocks (blue) and for the combined core blocks per junction (red).

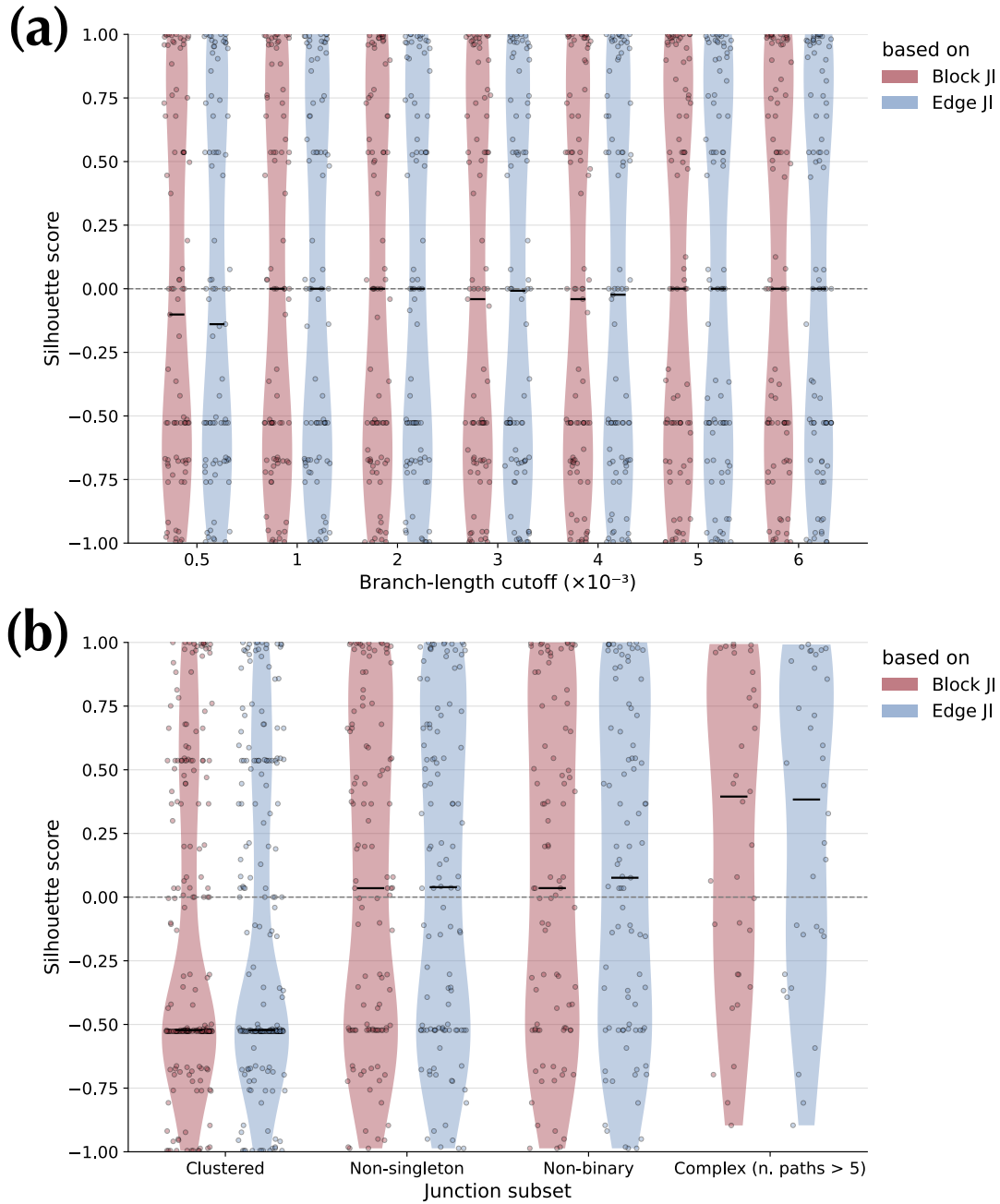


Figure 3: Per-junction silhouette scores assessing agreement between sequence-based clustering and block structure similarity. To evaluate how well clustering based on sequence similarity in flanking core blocks reflects block structure, we computed pairwise Jaccard indices between isolates using shared blocks and shared edges. For each junction, mean within-cluster and between-cluster similarities were used to derive silhouette scores. Values close to 1 indicate well-separated clusters, whereas values below 0 suggest greater similarity to isolates in other clusters than within the assigned cluster. (a) Silhouette scores are shown per junction across different branch-length cutoffs. Only junctions forming clusters under all tested thresholds are included. Scores remain largely stable across cutoffs, with median values around 0. The cutoff 5×10^{-4} shows slight deviations, consistent with being overly permissive. The observed bimodal distribution with high scores for some clusters and low scores for others, suggests that clustering often separates isolates with identical block structures, which is consistent with homologous recombination events that do not alter genomic content. (b) Subsetting silhouette scores at the final cutoff of 0.001 supports this interpretation. Singleton and binary junctions, which typically involve few structural changes, are associated with low silhouette scores.

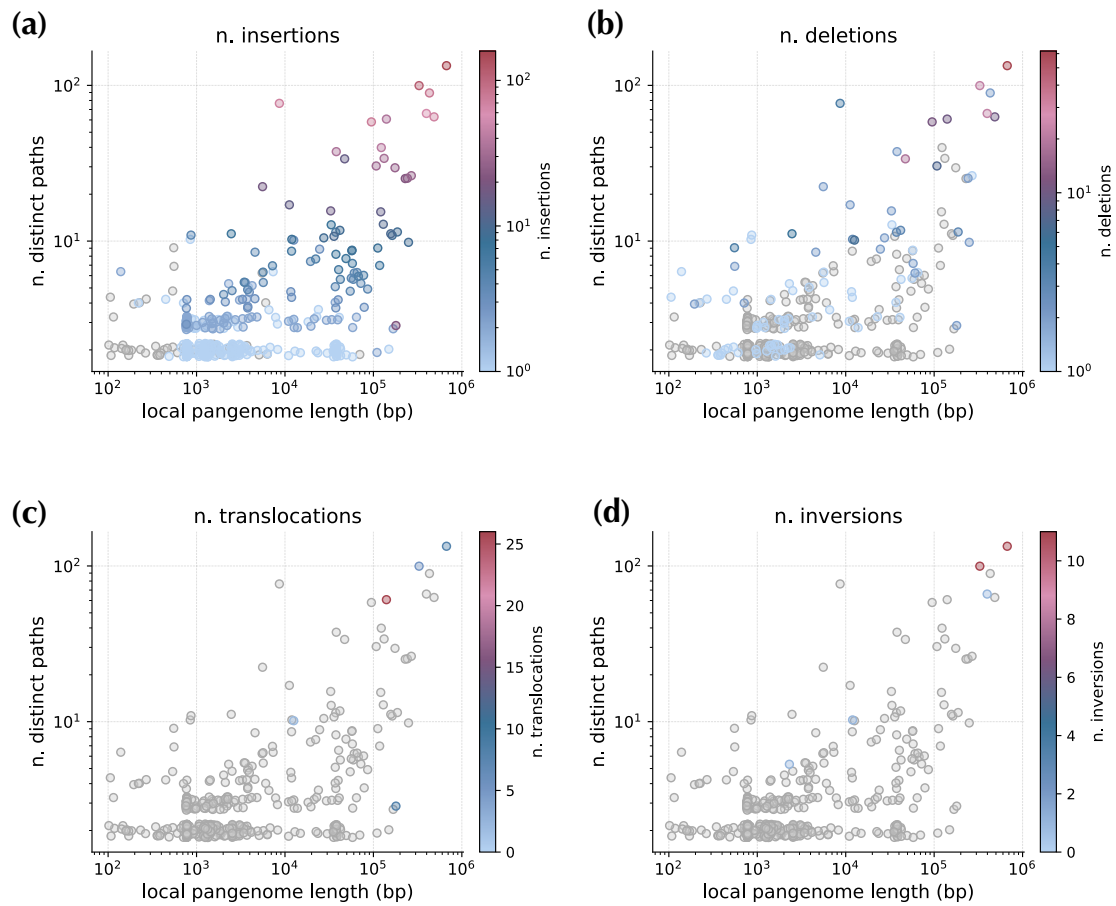


Figure 4: Associations between the number of structural event types and junction diversity. Relationship between the number of structural events detected per junction after clustering and junction diversity, quantified by local pangenome length and the number of distinct paths. Event types are shown separately for insertions (a), deletions (b), translocations (c), and inversions (d). For all types the number of events increases with junction diversity.

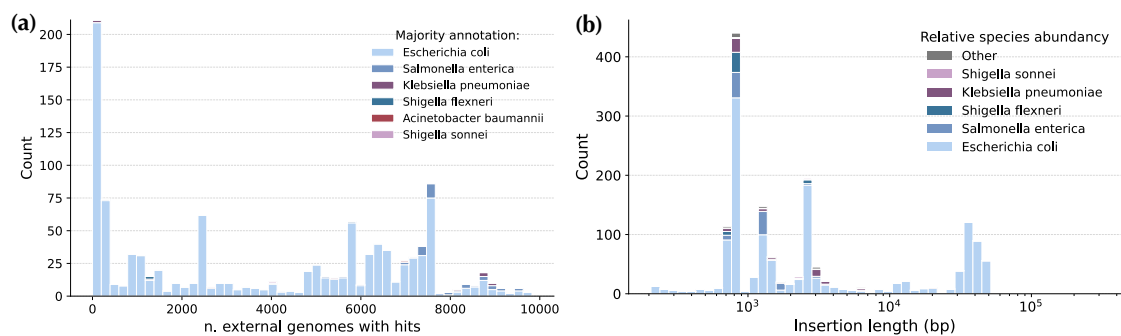


Figure 5: Species assignments by hit frequency and length. (a) Distribution of the number of external genome hits for detected insertions, annotated by their majority species. Non-*E. coli* majority assignments are observed primarily among insertions with high numbers of external hits. (b) Length distribution of detected insertions colored by species abundance. External species annotations are observed mainly for shorter insertions within the insertion-sequence peak.

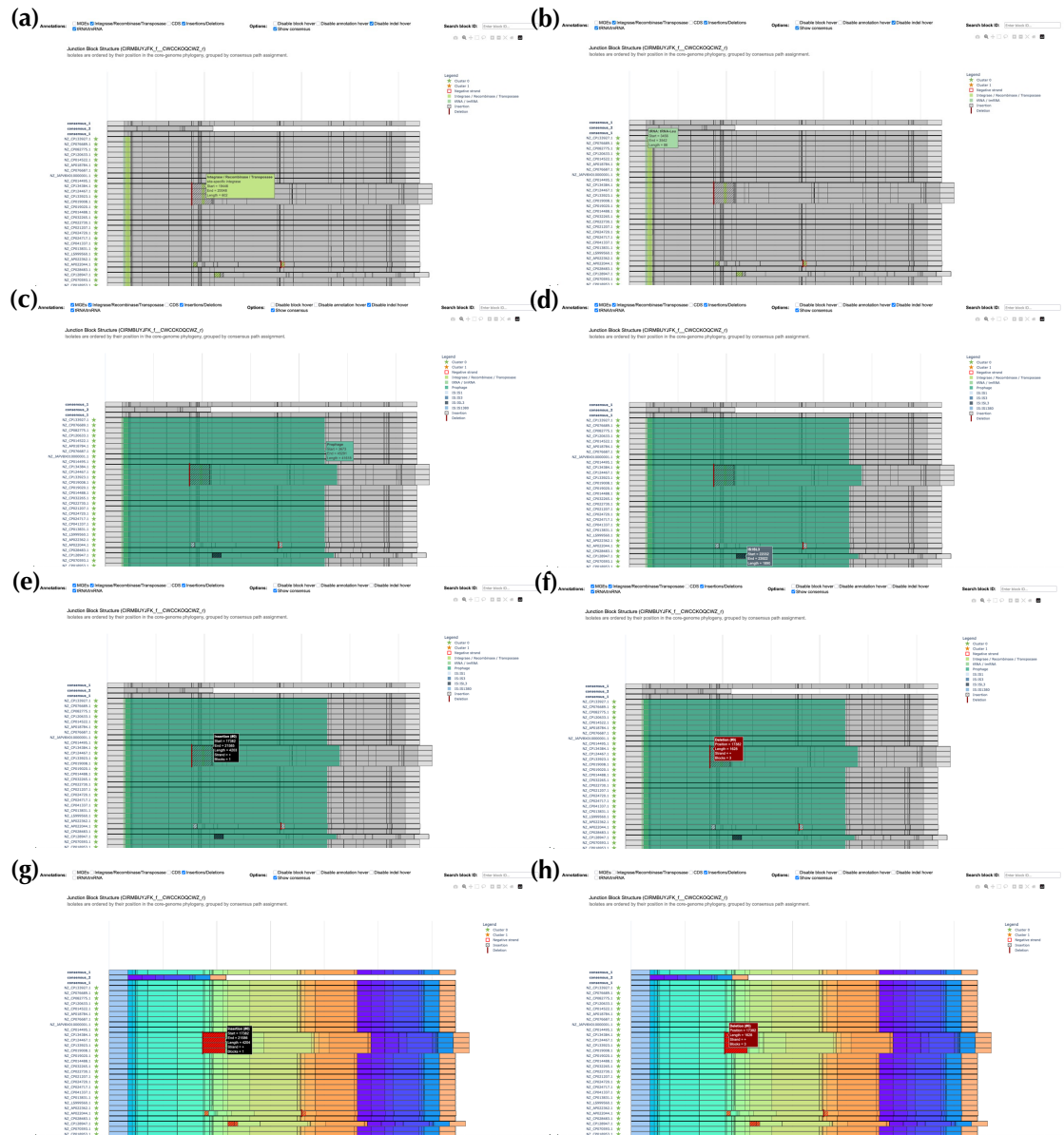


Figure 6: Hover-based visualization of genomic annotations and structural events in the interactive dashboard. Different hover views provide a quick overview of block structure and annotations. These include integrases, recombinases, and transposases (a); tRNA and tmRNA genes (b); mobile genetic elements, including prophages (c), insertion sequences and their family types (d); defense systems and integrons; and structural events, including insertions (e), deletions (f), translocations, and inversions. Structural events are also shown in block structure mode (g, h).

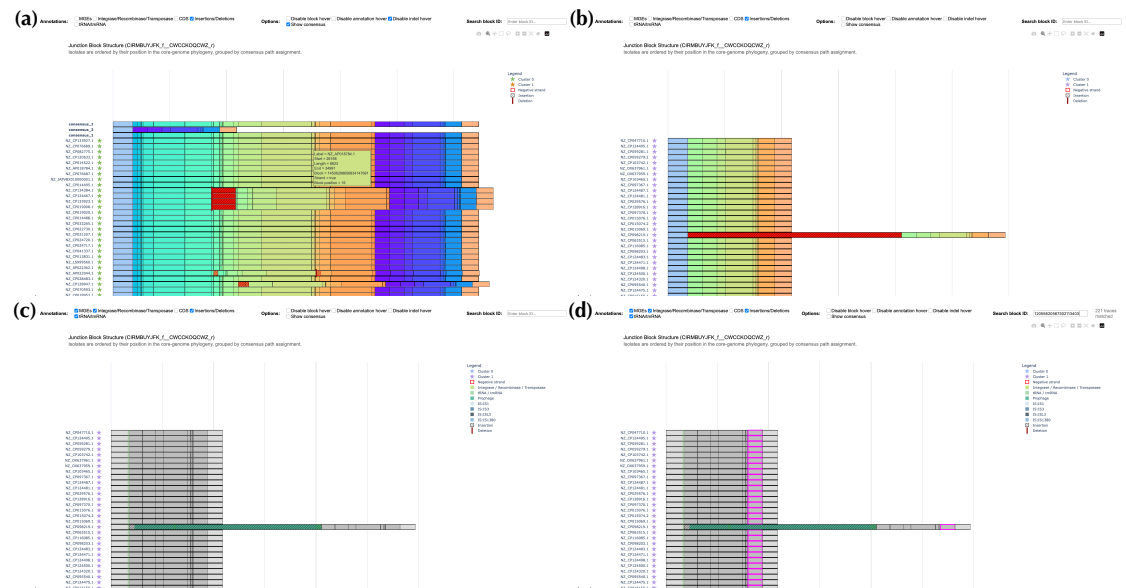


Figure 7: Additional dashboard features: block highlighting and core genome phylogenetic ordering. Within the dashboard, consensus path visualization can be toggled on (a) and off (b), allowing isolates to be displayed either within their clusters or according to the overall core genome phylogeny. A search field enables highlighting of blocks by block ID (c-d), which helps identify specific blocks and trace them across different isolates.

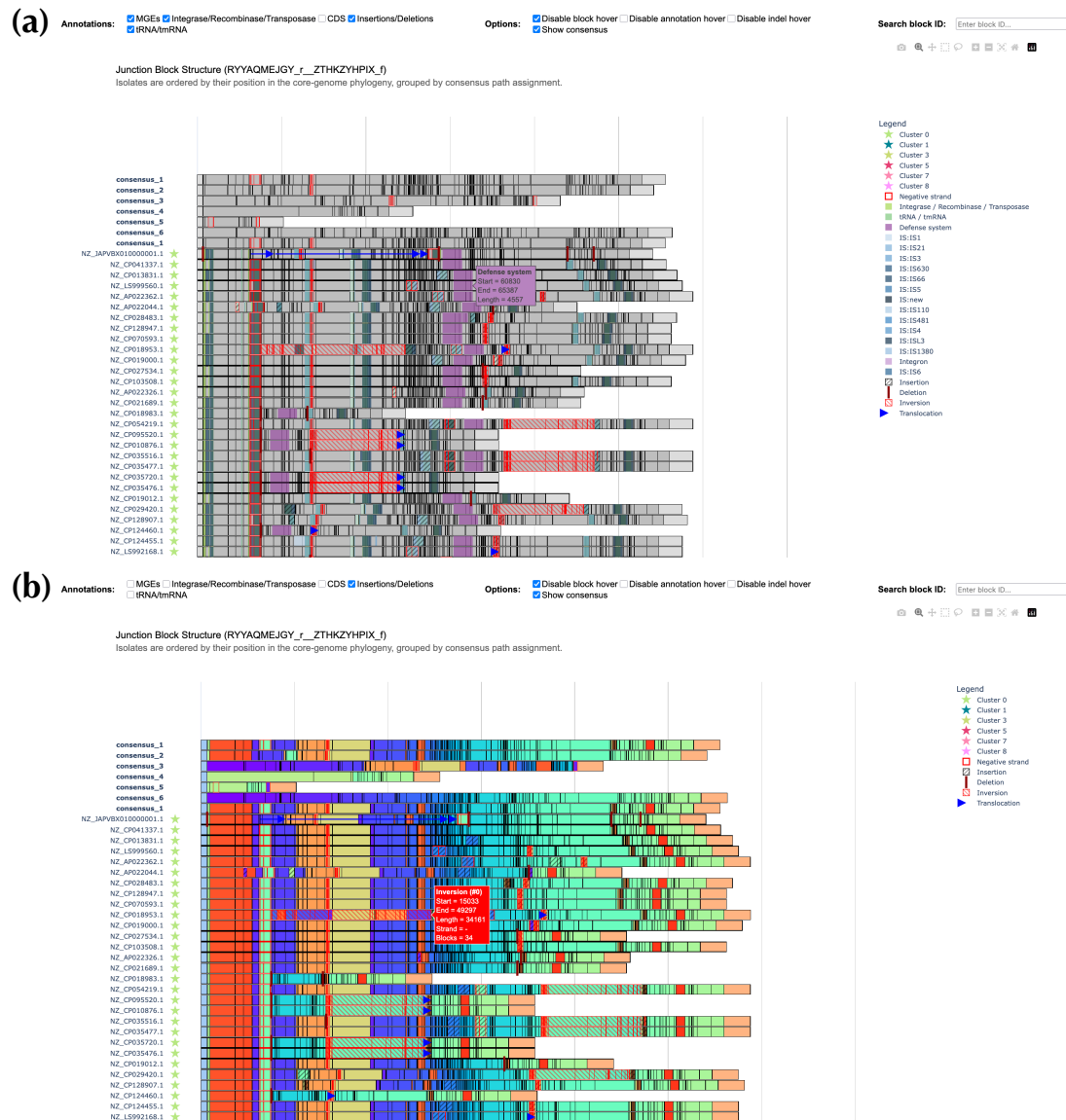


Figure 8: Example visualization of the high-complexity junction RYQAQMEJGY_r_ZTHKZYHPIX_f. High-complexity junctions are often difficult to interpret because of the multiple events that shape their structure. The dashboard provides an intuitive overview of block structure (a) and associated annotations (b). In this example, inversions (red diagonal lines) and translocations (blue arrows) are also visible.